



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 15052-001.0
Date: 9/10/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Concrete Encasement
Reviewed By: Mason Drapesa-Morse, Gavin Goo

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

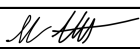
1. Provided shop drawings are incomplete and unreadable. **Shop drawings have been uploaded again**
2. Clearly mark key product data as specified in Section 1.04.B. For example, highlight areas where the proposed encasement to be used. **Proposed encasement areas are in bold.**

High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number:
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

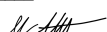
Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
SPECIFICATION SECTION # _____
SPECIFICATION _____
PARAGRAPH _____
DRAWING _____
SUBCONTRACTOR _____
SUPPLIER _____
MANUFACTURER _____

CERTIFIED BY CQCM or Designee : 

Specification section - Common
Work results for general piping

SECTION 15052

COMMON WORK RESULTS FOR GENERAL PIPING

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Basic materials and methods for metallic and plastic piping systems.

1.02 REFERENCES

- A. American Society of Mechanical Engineers (ASME):
 - 1. B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 Through 24.
 - 2. B16.47 - Large Diameter Steel Flanges: NPS 26 Through NPS 60 Metric/Inch Standard.
- B. American Water Works Association (AWWA):
 - 1. C11 - Rubber-Gasket Joints for Ductile-Iron Pressure Pipe.
 - 2. C105 - Polyethylene Encasement for Ductile-Iron Pipe Systems.
 - 3. C111 - Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
 - 4. C151 - Ductile-Iron Pipe, Centrifugally Cast.
 - 5. C207 - Standard for Steel Pipe Flanges for Waterworks Services-Size 4 In. Through 144 In.
- C. ASTM International (ASTM):
 - 1. A74 - Standard Specification for Cast Iron Soil Pipe and Fittings.
 - 2. A193 - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications.
 - 3. A194 - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both.
 - 4. A307 - Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60,000 PSI Tensile Strength.
 - 5. A320 - Standard Specification for Alloy-Steel and Stainless Steel Bolting for Low-Temperature Service.
 - 6. A563 - Standard Specification for Carbon and Alloy Steel Nuts.
 - 7. B88 - Standard Specification for Seamless Copper Water Tube.
 - 8. F37 - Standard Test Methods for Sealability of Gasket Materials.
 - 9. F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
 - 10. F594 - Standard Specification for Stainless Steel Nuts.
 - 11. F2329 - Standard Specification for Zinc Coating, Hot-Dip, Requirements of Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners.

- D. California Health and Safety Code.
- E. NSF International (NSF):
 - 1. 61 – Drinking Water System Components – Health. Effects.
 - 2. 372 - Drinking Water System Components - Lead Content.

1.03 DEFINITIONS

- ✓ A. Buried pipes: Pipes that are buried in the soil with or without a concrete pipe encasement.
- B. Exposed pipe: Pipes that are located above ground, or located inside a structure, supported by a structure, or cast into a concrete structure.
- C. Underground pipes: Buried pipes - see A. above.
- D. Underwater pipes: Pipes below the top of walls in basins or tanks containing water.
- E. Wet wall: A wall with water on at least 1 side.
- F. Pipes adjacent to a wet wall: Pipe centerline within 10 inches of the wet wall.

1.04 SUBMITTALS

- A. Submit as specified in Section 01330 - Submittal Procedures.
- ✓ B. Product data:
 - 1. For each piping product in this Section as applicable:
 - a. Design features.
 - b. Load capacities.
 - c. Material designations by UNS alloy number or ASTM Specification and Grade.
 - d. Data needed to verify compliance with the Specifications.
 - e. Catalog data.
 - f. Clearly mark submittal information to show specific items, materials, and accessories or options being furnished.
 - ✓ g. Detailed piping layout drawings.
- C. Calculations:
 - 1. Provide calculations in accordance with NSF 372 for materials in contact with drinking water.

PART 2 PRODUCTS

2.01 GENERAL

- A. Pipes:
 - 1. Provide new pipe.

2. Piping provided from manufacturers/distributors inventory is subject to the following condition:
 - a. Provide proof pipe manufactured more than 6 months prior to delivery was stored properly and the material and/or coating was not subjected to ultraviolet (UV) degradation.
3. Mark each piece of pipe in accordance with applicable standards.

2.02 LINK TYPE SEALS

- A. Characteristics:
 1. Modular mechanical type, consisting of interlocking neoprene or synthetic rubber links shaped to continuously fill the annular space between the pipe and wall opening.
 2. Links to form a continuous rubber belt around the pipe.
 3. Provide a nylon polymer pressure plate with Type 316 stainless steel hardware. Isolate pressure plate from contact with wall sleeve.
 4. Hardware to be Type 316 stainless steel.
 - a. Provide anti-galling lubricant for threads.
- B. One of the following or equal:
 1. Link-Seal.
 2. Pipe Linx.

2.03 FLANGE BOLTS AND NUTS

- A. General:
 1. Washer:
 - a. Provide a washer for each nut.
 - b. Washer shall be of the same material as the nut.
 2. Nuts: Heavy hex-head.
 - a. Stainless steel nuts shall be finished with TRIPAC 2000 coating system to minimize galling and provide proper torque.
 3. Cut and finish flange bolts to project:
 - a. Face of the bolt shall exceed face of nut by 1/8 inch minimum.
 - b. A maximum of 1/4 inch beyond outside face of nut after assembly.
 4. Tap holes for cap screws or stud bolts when used.
 5. Lubricant for stainless steel bolts and nuts:
 - a. Chloride-free.
 - b. Manufacturers: One of the following or equal:
 - 1) Huskey FG-1800 Anti-Seize.
 - 2) Weicon Anti-Seize High-Tech.
- B. Ductile iron pipe:
 1. On exposed pipes with pressures equal to or less than 150 pounds per square inch gauge (psig):
 - a. Bolts: In accordance with ASTM A307, Grade B.
 - b. Nuts: In accordance with ASTM A563, Grade A.
 - c. Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 2. On exposed pipes with pressures greater than 150 psig:
 - a. Bolts: In accordance with ASTM A193, Grade B7.

- b. Nuts: In accordance with ASTM A194, Grade 2H.
 - c. Bolts and nuts: Hot-dip galvanized in accordance with ASTM F2329.
- 3. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
- 4. On buried pipes:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M for nuts.
 - c. Encase in 2 layers of loose polyethylene wrap in accordance with AWWA C105.
- C. Plastic pipe:
 - 1. On exposed pipes:
 - a. Bolts: In accordance with ASTM A307, Grade B.
 - b. Nuts: In accordance with ASTM A563, Grade A.
 - c. Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 - 2. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
- D. Steel pipe:
 - 1. On exposed pipes:
 - a. For ASME B16.5 Class 150 flanges and AWWA C207 Class D flanges:
 - 1) Bolts: In accordance with ASTM A307, Grade B.
 - 2) Nuts: In accordance with ASTM A563, Grade A.
 - 3) Bolts and Nuts: Hot-dip galvanized in accordance with ASTM F2329.
 - b. For ASME B16.5 and B16.47 Class 300 flanges and AWWA C207 Class E and F flanges:
 - 1) Bolts: In accordance with ASTM A193, Grade B7.
 - 2) Nuts: In accordance with ASTM A194, Grade 2H.
 - 2. On underwater pipes and pipes adjacent to wet walls:
 - a. Bolts: In accordance with ASTM A193, Grade B8M.
 - b. Nuts: In accordance with ASTM A194, Grade 8M.
 - 3. Low-temperature service: Stainless steel, Type 316:
 - a. Bolts: In accordance with ASTM A320, Grade B8M, Class 1, heavy hex.
 - b. Nuts: In accordance with ASTM A194, Grade B8M, heavy hex.
 - c. Washers: Alloy group matching bolts and nuts.
 - 4. High-temperature service or high-pressure service: Stainless steel, Type 316:
 - a. Bolts: In accordance with ASTM A193, Grade B8M, Class 1, heavy hex.
 - b. Nuts: In accordance with ASTM A194, Grade 8, heavy hex.
 - c. Washers: Alloy group matching bolts and nuts.

2.04 MECHANICAL JOINTS BOLTS AND NUTS

- A. Bolts including T-Bolts:
 - 1. Type 316 stainless steel in accordance with ASTM F593.
- B. Heavy Hex Nuts:
 - 1. Type 316 stainless steel in accordance with ASTM F594, finished with TRIPAC 2000 coating system to minimize galling and provide proper torque.

2.05 GASKETS

- A. General.
 - 1. Gaskets shall be suitable for the specific fluids, pressure, and temperature conditions.
 - 2. Capable of being applied on surface of piping with cavities to provide for an improved seal with the internal piping pressure.
- B. Gaskets for flanged joints in ductile iron and steel piping for sewage service:
 - 1. Suitable for pressures equal and less than 150 pounds per square inch gauge, temperatures equal and less than 250 degrees Fahrenheit, and raw sewage service.
 - 2. Gasket material:
 - a. SBR or neoprene elastomer with minimum Shore A hardness value of 70.
 - b. Reinforcement: Cloth or synthetic fiber.
 - c. Thickness: Minimum 3/32-inch thick for less than 10-inch pipe; minimum 1/8 inch thick for 10-inch and larger pipe.
 - 3. Manufacturers: One of the following or equal:
 - a. Pipe less than 48 inches in diameter:
 - 1) Garlock, Style 7797.
 - 2) John Crane, similar product.
 - b. Pipe 48 inches in diameter and larger:
 - 1) Garlock, Style 3760.
 - 2) John Crane, similar product.
- C. Gaskets for grooved end ductile iron and steel piping:
 - 1. Suitable for pressures equal to the encapsulating coupling or flange adapter.
 - 2. Material: Pressure responsive elastomer.
 - a. Ductile iron piping: FlushSeal® type.
 - 1) Halogenated Butyl: Grade M; for temperatures to 200 degrees Fahrenheit.
 - 2) Nitrile: Grade S; for temperatures to 180 degrees Fahrenheit.
 - b. Steel piping:
 - 1) EPDM: Grade E; for temperatures to 230 degrees Fahrenheit.
 - 2) EPDM-HP: Grade EHP, for temperatures to 250 degrees Fahrenheit.
 - 3) Nitrile: Grade T, for temperatures to 180 degrees Fahrenheit.
 - 3. Gaskets shall be verified as suitable for the intended service.
 - a. Temperature ratings may vary depending on the fluid/media.
 - 4. Manufacturers: Gaskets shall be of the same manufacturer as the encapsulating couplings/flange adapters.
 - a. Victaulic Company.
- D. Gaskets for flanged joints in polyvinyl chloride and polyethylene piping:
 - 1. Suitable for pressures equal to and less than 150 pounds per square inch gauge, with low flange bolt loadings, temperatures equal to and less than 120 degrees Fahrenheit, and polymer, chlorine, caustic solutions, and other chemicals, except chemicals which liberate free fluorine including fluorochemicals and gaseous fluorine.
 - 2. Material:
 - a. Chemical systems: 0.125-inch thick Viton™ rubber.
 - b. Sewer and water: 0.125-inch thick SBR.

3. Manufacturers: One of the following or equal:
 - a. Garlock.
 - b. John Crane, similar product.
- E. Gaskets for flanged joints in gas or liquefied petroleum gas piping:
 1. Digester gas in stainless steel, or black steel piping: Suitable for pressures equal to and less than 150 pounds per square inch gauge, temperatures equal to and less than 200 degrees Fahrenheit, and digester gas and mild acid concentrations.
 2. Chlorine gas application in black steel piping: Suitable for pressures equal to or less than 300 pounds per square inch gauge, temperatures equal to or less than 100 degrees Fahrenheit, and chlorine gas application.
 3. Liquefied petroleum, propane, and natural gas applications in black steel piping: Suitable for pressures equal to and less than 250 pounds per square inch gauge, temperatures equal to and less than 100 degrees Fahrenheit, and liquefied petroleum gas, propane gas, and natural gas application.
 4. Material:
 - a. Microcellular Teflon™ outer layers with rigid center layer.
 - b. Sealability in accordance with ASTM F37, less than 0.55 millimeters per hour leakage of iso-octane at 1,000 pounds per square inch gasket load and 9.8 pounds per square inch fluid pressure.
 5. Manufacturers: One of the following or equal:
 - a. Garlock, Style 3545.
 - b. John Crane, similar product.
- F. Gaskets for flanged joints in low pressure air piping:
 1. Suitable for pressures equal to and less than 150 pounds per square inch gauge, temperatures equal to and less than 300 degrees Fahrenheit, and compressed air service.
 2. Material: EPDM elastomer, 1/8-inch thick, 60 Shore hardness, smooth surface.
 3. Manufacturers: One of the following or equal:
 - a. Garlock, Style 8314.
 - b. John Crane, similar product.
- G. Gaskets for flanged joints in ductile iron or steel water piping:
 1. Suitable for hot or cold water, pressures equal to and less than 150 pounds per square inch gauge, and temperatures equal to and less than 160 degrees Fahrenheit.
 2. Material:
 - a. SBR or neoprene elastomer, compressed, with non-asbestos fiber reinforcement.
 3. Manufacturers: One of the following or equal:
 - a. Garlock, Bluegard 3300.
 - b. John Crane, similar product.
- H. Gaskets for flanged joints in ductile iron or steel drinking water piping meeting NSF 61 requirements:
 1. Suitable for hot or cold water, pressures equal to or less than 150 pounds per square inch gauge, and temperatures equal to or less than 160 degrees Fahrenheit.

2. Material:
 - a. EPDM material with 80 Shore A durometer rating.
3. Manufacturers: One of the following or equal:
 - a. Garlock, 98206.
 - b. John Crane, similar product.

PART 3 EXECUTION

3.01 INSTALLATION

- A. General:
 1. Piping drawings:
 - a. Except in details, piping is indicated diagrammatically. Not every offset and fitting, or structural difficulty that may be encountered has been indicated on the Drawings. Sizes and locations are indicated on the Drawings.
 - b. Perform minor modifications to piping alignment where necessary to avoid structural, mechanical, or other type of obstructions that cannot be removed or changed.
 - 1) Modifications are intended to be of minor scope, not involving a change to the design concept or a change to the Contract Price or Contract Times.
 2. Piping alternatives:
 - a. Provide piping as specified in this Section, unless indicated on the Drawings or specified otherwise.
 - b. Alternative pipe ratings:
 - 1) Piping with greater pressure rating than specified may be substituted in lieu of specified piping without changes to the Contract Price.
 - 2) Piping of different material may not be substituted in lieu of specified piping.
 - c. Valves in piping sections: Capable of withstanding specified test pressures for piping sections and fabricated with ends to fit piping.
 - d. Grooved joints: Use couplings, flange adapters, and fittings of the same manufacturer.
 - 1) Manufacturer's factory trained representative:
 - a) Provide on-site training for Contractor's field personnel.
 - b) Periodically visit the jobsite to verify Contractor is following best recommended practices.
 - 2) Distributor's representative is not considered qualified to conduct the training or jobsite visits.
 - e. Flanged joints: where 1 of the joining flanges is raised face type, provide a matching raised face type flange for the other joining flange.
 - 1) Wire brush and clean flange faces to remove all oil, grease, loose primer, mill scale, and any other foreign matter that could affect the proper seating of the gasket.
 3. Unless otherwise indicated on the Drawings, piping at pipe joints, fittings, couplings, and equipment shall be installed without rotation, angular deflection, vertical offset, or horizontal offset.

- B. Wall and slab penetrations:
1. Provide sleeves for piping penetrations through aboveground masonry and concrete walls, floors, ceilings, roofs, unless specified or otherwise indicated on the Drawings.
 2. For piping 1 inch in nominal diameter and larger, provide sleeves with minimum inside diameters of 1 inch plus outside diameter of piping. For piping smaller than 1 inch in nominal diameter, provide sleeve of minimum twice the outside diameter of piping.
 - a. Arrange sleeves and adjacent joints so piping can be pulled out of sleeves and replaced without disturbing the structure.
 - b. Cut ends of sleeves flush with surfaces of concrete, masonry, or plaster.
 - c. Conceal ends of sleeves with escutcheons where piping runs through floors, walls, or ceilings of finished spaces within buildings.
 - d. Seal spaces between pipes and sleeves with link-type seals when not otherwise specified or indicated on the Drawings.
 3. Provide flexibility in piping connecting to structures to accommodate movement due to soil settlement and earthquakes. Provide flexibility using details indicated on the Drawings.
 4. Core drilled openings:
 - a. Do not damage or cut existing reinforcing bars, electrical conduits, or other items embedded in the existing concrete without acceptance by Engineer.
 - b. Determine location of reinforcing bars or other obstructions with a non-destructive indicator device.
 - c. Remove dust and debris from hole using compressed air.
- C. Exposed piping:
1. Install exposed piping in straight runs parallel to the axes of structures, unless otherwise indicated on the Drawings:
 - a. Install piping runs plumb and level, unless otherwise indicated on the Drawings.
 - 1) Slope plumbing drain piping with a minimum of 1/4-inch per foot downward in the direction of flow.
 2. Install exposed piping after installing equipment and after piping and fitting locations have been determined.
 3. Support piping: As specified in Sections 15061 - Pipe Supports, 15062 - Preformed Channel Pipe Support System, and 15063 - Non-Metallic Pipe Support System:
 - a. Do not transfer pipe loads and strain to equipment.
 4. In addition to the joints indicated on the Drawings, provide unions, flexible couplings, flanged joints, flanged coupling adapters, and other types of joints or means which are compatible with and suitable for the piping system, and necessary to allow ready assembly and disassembly of the piping.
 5. Assemble piping without distortion or stresses caused by misalignment:
 - a. Match and properly orient flanges, unions, flexible couplings, and other connections.
 - b. Do not subject piping to bending or other undue stresses when fitting piping.
 - c. Do not correct defective orientation or alignment by distorting flanged joints or subjecting flange bolts to bending or other undue stresses.

- d. Flange bolts, union halves, flexible connectors, and other connection elements shall slip freely into place.
- e. Alter piping assembly to fit when proper fit is not obtained.
- f. Install eccentric reducers or increasers with the top horizontal for pump suction piping.

D. Buried piping:

- 1. Bury piping with minimum 3-foot cover without air traps, unless otherwise indicated on the Drawings.
- 2. Where 2 similar services run parallel to each other, piping for such services may be laid in the same trench.
 - a. Lay piping with sufficient room for assembly and disassembly of joints, for thrust blocks, for other structures, and to meet separation requirements of public health authorities having jurisdiction.
- 3. Laying piping:
 - a. Lay piping in finished trenches free from water or debris. Begin at the lowest point with bell ends up slope.
 - b. Place piping with top or bottom markings with markings in proper position.
 - c. Lay piping on an unyielding foundation with uniform bearing under the full length of barrels.
 - d. Where joints require external grouting, banding, or pointing, provide space under and immediately in front of the bell end of each section laid with sufficient shape and size for grouting, banding, or pointing of joints.
 - e. At the end of each day's construction, plug open ends of piping temporarily to prevent entrance of debris or animals.
- ✓ 4. Concrete encase buried pipe installed under concrete slabs or structures.

E. Venting piping under pressure:

- 1. Lay piping under pressure flat or at a continuous slope without air traps, unless otherwise indicated on the Drawings.
- 2. Install plug valves as air bleeder cocks at high points in piping.
 - a. Provide 1-inch plug valves for water lines, and 2-inch plug valves for sewage and sludge lines, unless otherwise indicated on the Drawings.
- 3. Provide additional pipe taps with plug cocks and riser pipes along piping as required for venting during initial filling, disinfecting, and sampling.
- 4. Before piping is placed into service, close plug valves and install plugs. Protect plugs and plug valves from corrosion in as specified in Section 09960 - High-Performance Coatings.

F. Condensate in digester gas piping:

- 1. Slope digester gas piping to drip traps or low-point drains at minimum 1/2-inch per foot where condensate flows against the gas or 1/4-inch per foot where condensate flows with gas.
- 2. Install tapered filler pieces between flanges at high points of straight runs to provide for slope reversals.
 - a. Do not subject piping to high stresses in order to change direction.
- 3. Provide pipe taps, threaded nipples, and 1-inch plug valves at low points in concrete utility boxes with lids.

- G. Restraining buried piping:
1. Restrain piping at valves and at fittings where piping changes direction, changes sizes, and at ends:
 - a. When piping is underground, use mechanical restraints or push-on restraints.
 - 1) Concrete thrust blocks shall not be used unless specifically called out with a note on the Drawings or specified.
 - b. Determine thrust forces by multiplying the nominal cross-sectional area of the piping by design test pressure of the piping.
 2. Provide restraints with ample size to withstand thrust forces resulting from test pressures:
 - a. During testing, provide suitable temporary restraints where piping does not require permanent restraints.
 3. Where thrust blocks are allowed, place concrete thrust blocks against undisturbed soil.
 4. Where thrust blocks are allowed, place concrete so piping joints, fittings, and other appurtenances are accessible for assembly and disassembly.
 5. Provide underground mechanical restraints where specified in Attachment A - Piping Schedule.
- H. Restraining above ground and underwater piping:
1. Restrain piping at valves and at fittings where piping changes direction, changes sizes, and at ends:
 - a. When piping is aboveground or underwater, use mechanical or structural restraints.
 - b. Determine thrust forces by multiplying the nominal cross-sectional area of the piping by design test pressure of the piping.
 2. Provide restraints with ample size to withstand thrust forces resulting from test pressures:
 - a. During testing, provide suitable temporary restraints where piping does not require permanent restraints.
- I. Connections to existing piping:
1. Expose existing piping to which connections are to be made with sufficient time to permit, where necessary, field adjustments in line, grade, or fittings:
 - a. Protect domestic water/potable water supplies from contamination:
 - 1) Make connections between domestic water supply and other water systems in accordance with requirements of public health authorities.
 - 2) Provide devices approved by Owner of domestic water supply system to prevent flow from other sources into the domestic supply system.
 2. Make connections to existing piping and valves after sections of new piping to be connected have been tested and found satisfactory.
 3. Provide sleeves, flanges, nipples, couplings, adapters, and other fittings needed to install or attach new fittings to existing piping and to make connections to existing piping.
 4. For flanged connections, provide stainless steel bolts with isolation bushings and washers, and full-face flange gaskets.

- J. Connections between ferrous and nonferrous metals:
 - 1. Connect ferrous and nonferrous metal piping, tubing, and fittings with dielectric couplings especially designed for the prevention of chemical reactions between dissimilar metals.
 - 2. Nonferrous metals include aluminum, copper, and copper alloys.
- K. Flanged connections between dissimilar metals such as ductile iron pipe and steel pipe:
 - 1. Provide stainless steel bolts with isolation bushings and washers, and full-face flange gaskets.
- L. Where stainless steel bolts and/or nuts are in contact with non-stainless steel metals, use isolation bushings and washers.

3.02 CLEANING

- A. Piping cleaning:
 - 1. Upon completion of installation, clean piping interior of foreign matter and debris.
 - 2. Perform special cleaning when required by the Contract Documents.
- B. Cleaning potable water piping:
 - 1. Flush and disinfect potable water piping as specified in Section 01757 - Disinfection, and other regulatory requirements.
- C. Cleaning air piping:
 - 1. Perform special cleaning of filtered air piping from the intake clean air plenums to the discharge points and high-pressure air piping.
 - a. Protect surfaces from contamination.
 - 2. Special cleaning shall include wire brushing, power tool cleaning, wiping down with lint-free cloths, brooming, and vacuuming to remove rust, scale, weld spatter, dust, dirt, oil, and other matter deleterious to operation of the air system:
 - a. Do not sandblast installed piping.
 - 3. To the greatest extent possible, clean piping immediately prior to final closure of piping systems:
 - a. Enter piping, clean and wipe down surfaces, and vacuum out residue.
 - b. Clean surfaces not accessible to this cleaning operation after installation within 6 hours preceding installation.
 - 4. Subsequent to cleaning, protect surfaces from contamination by dust, dirt, construction debris, and moisture, including atmospheric moisture:
 - a. Whether or not pipe upstream has been cleaned, temporarily seal openings in partially completed work except when installation is actively in progress.
 - b. When installation is actively in progress, seal openings at the end of each day's construction or when construction is temporarily stopped.
 - 5. Suspend cleaning and seal openings when inclement weather, including dust storms, is imminent.
 - 6. Use clean, dry air for testing the piping and other elements of the system.

7. Prior to introduction of air to the system, blow piping clean.
 - a. Blow with maximum discharge rate possible for minimum 4 hours, using new blowers or compressors and filters.
 8. Clean surfaces that become contaminated prior to acceptance.
- D. Cleaning steam piping:
1. Thoroughly clean steam piping, boiler feed water piping, and condensate return piping by flushing with water.
 - a. Conduct cleaning operation after piping connections are completed and after piping has successfully passed pressure test.
 2. Utilize potable quality water, softened to remove hardness, for flushing water.
 - a. Provide temporary facilities and chemicals to add corrosion control additives to flushing water.
 - b. Utilize recommendations of boiler manufacturer for type and quantity of corrosion control additives.
 - c. When corrosion control additives are to be disposed of within treatment plant, use additives that are non-toxic to the treatment process.
 3. When the boiler system includes a feed water softening system, this system may be utilized as source of flushing water.
 - a. Provide adequately sized temporary storage tank when capacity of feed water softening system is less than required flow rate for flushing.
 4. Flush piping with water in sufficient quantities to produce a flow velocity of 6 feet per second in the main pipe and branch lines.
 5. Continue flushing until at least twice the volume of pipe has been flushed.
 6. Disconnect steam traps and piping less than 1/2-inch in nominal diameter prior to flushing operation.
 - a. After flushing operation is completed, drain entire system, and reconnect piping and steam traps.
 7. When cleaning operation is complete, remove temporary piping, pumps, tanks, and chemicals used to accomplish pipe cleaning.
 8. Prior to beginning cleaning operation, submit plan for the cleaning operation. Include details of equipment, procedures, and MSDS sheets for chemicals to be utilized.
- E. Conduct pressure and leak test, as specified.

3.03 COMMISSIONING

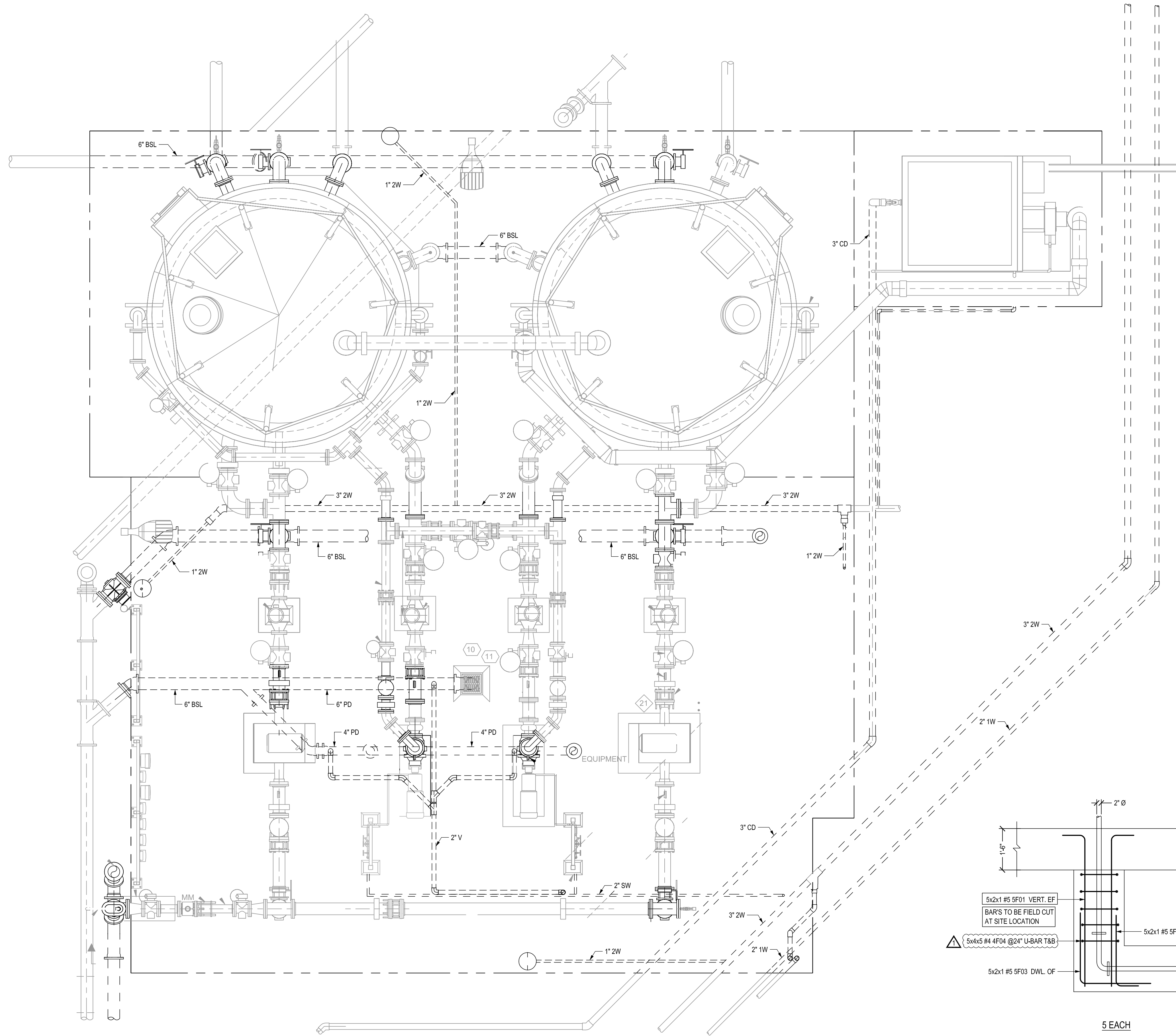
- A. As specified in Section 01756 - Commissioning and this Section.
- B. Field testing:
1. Piping system:
 - a. Witnessed.
 - b. Conduct pressure and leak test, as specified.

3.04 PIPING SCHEDULE

- A. Provide piping As specified in Attachment A - Piping Schedule unless noted otherwise in the Specifications or indicated on the Drawings with a written callout or note.

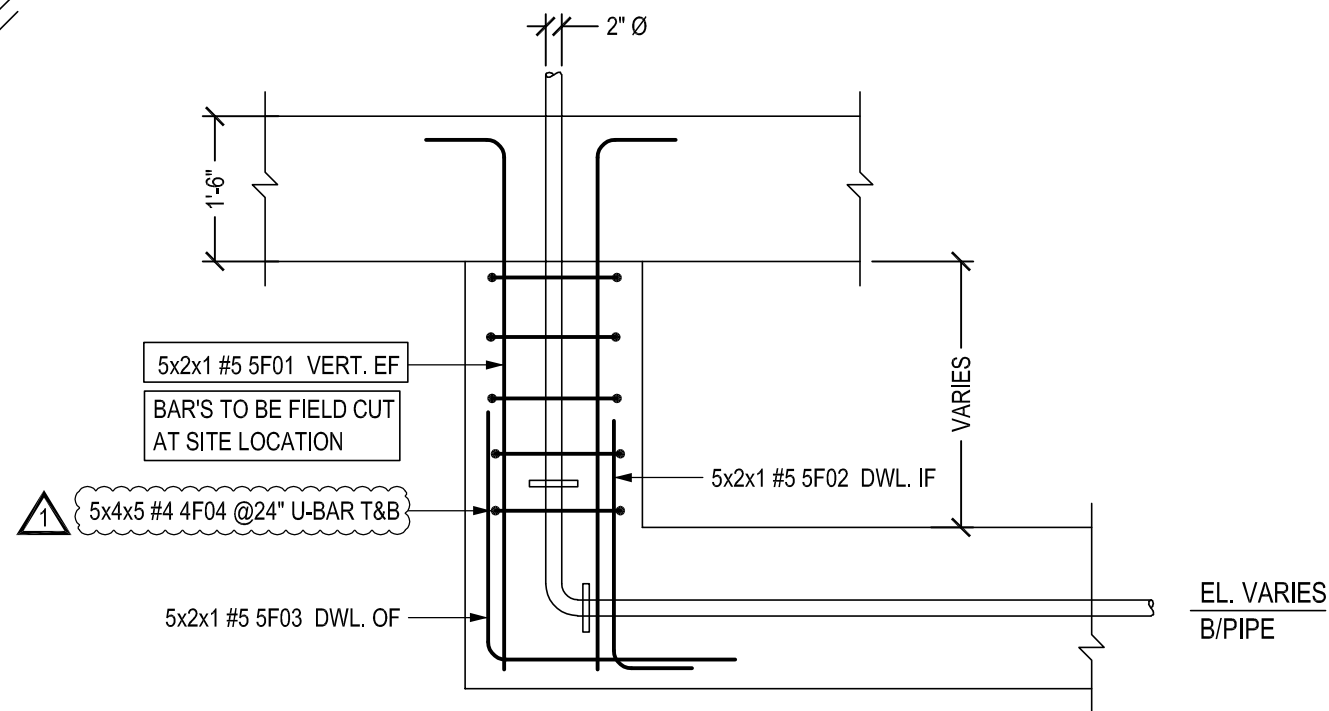
END OF SECTION

Concrete Encasement - Shop drawings



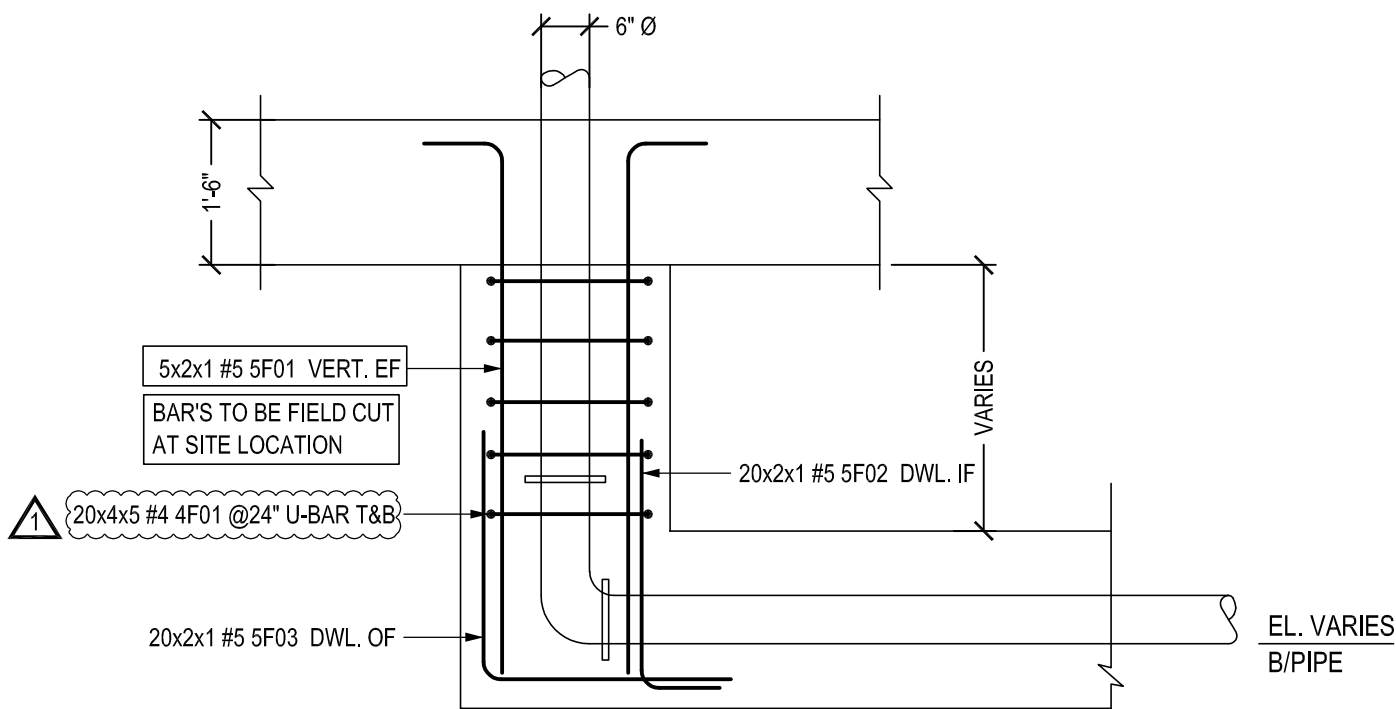
SLUDGE BLENDING BUILDING - PIPE ENCASEMENT LAYOUT

SCALE - 3/8"=1'-0"
REF:10-M-01-101



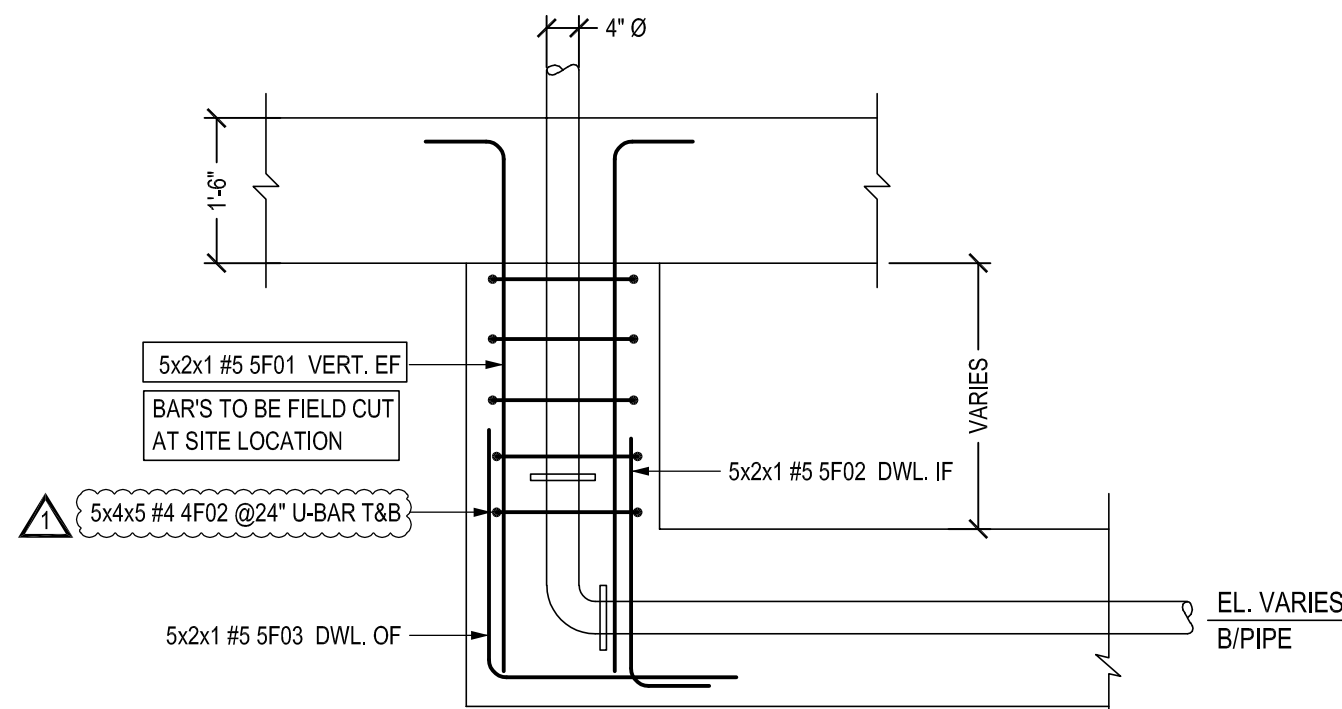
TYPICAL PIPE ENCASEMENT REINFORCEMENT DETAILS AT SLAB

SCALE - 1/2"=1'-0"
REF:00-T-05-702



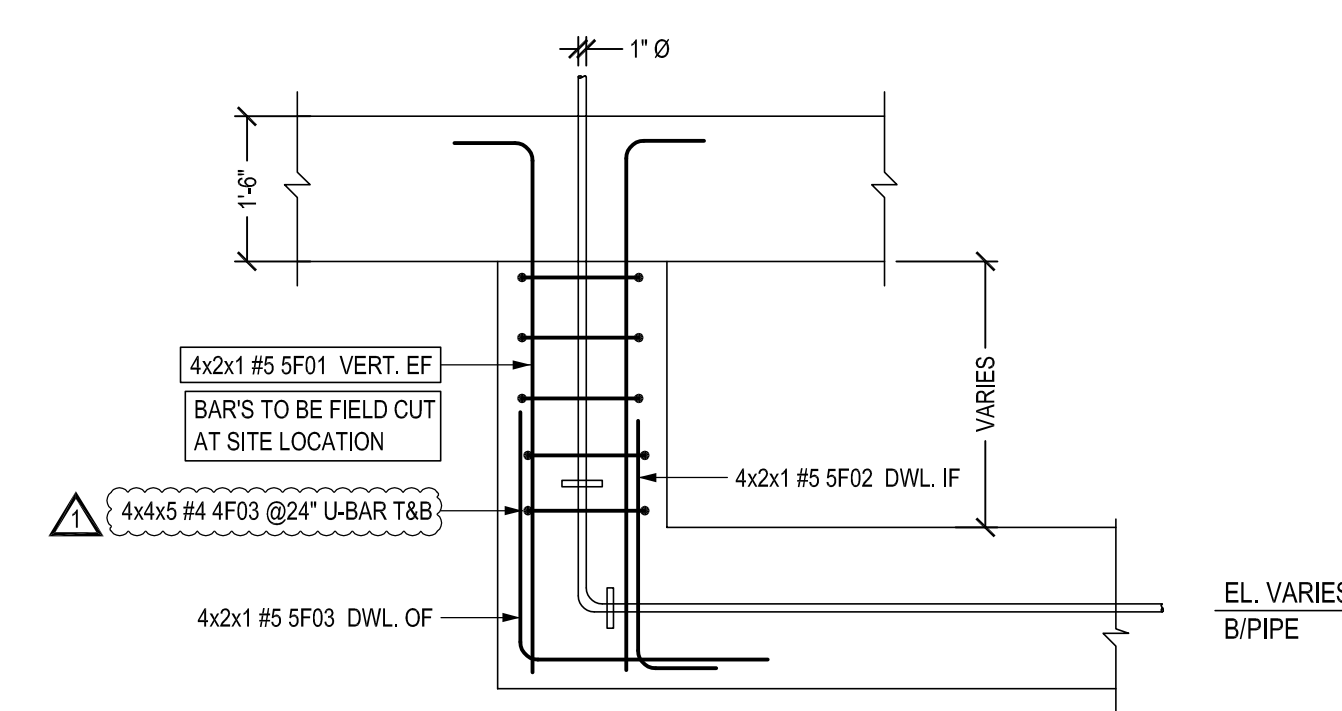
TYPICAL PIPE ENCASEMENT REINFORCEMENT DETAILS AT SLAB

SCALE - 1/2"=1'-0"
REF:00-T-05-702



TYPICAL PIPE ENCASEMENT REINFORCEMENT DETAILS AT SLAB

SCALE - 1/2"=1'-0"
REF:00-T-05-702



TYPICAL PIPE ENCASEMENT REINFORCEMENT DETAILS AT SLAB

SCALE - 1/2"=1'-0"
REF:00-T-05-702

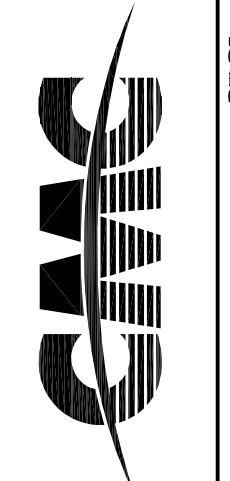
FIELD USE:
THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND CONSTRUCTION OF THE REINFORCEMENT. IT IS THE RESPONSIBILITY OF THE FIELD PERSONNEL TO VERIFY THE LOCATION AND DEPTH OF THE REINFORCEMENT. ANY DISCREPANCIES SHOULD BE NOTED AT THE TIME OF APPROVAL. THE FIELD PERSONNEL SHALL BE RESPONSIBLE FOR THE LAYOUT OF THE BUILDING OR FORMS.

APPROVAL AUTHORITY:
ALL ITEMS THAT ARE CIRCLED OR MARKED "VERIFY" REFER TO QUESTIONS, OMISSIONS, CONFLICTS OR ERRORS. REFER TO QUESTIONS, OMISSIONS, CONFLICTS OR ERRORS. ACTING IS INDICATED ON THIS APPROVAL. ALL ITEMS SO MARKED WILL BE ASSIGNED CORRECTIONS AS SUBMITTED TO THE FIELD PERSONNEL. THE FIELD PERSONNEL SHALL BE RESPONSIBLE FOR THE LAYOUT OF THE BUILDING OR FORMS. REQUESTED IN CONJUNCTION.

SUBMITTAL RECORD
DATE: 06/13/2025 FOR: APPROVAL
DATE: FOR:
DATE: 08/13/25 - REVISED AS PER MAIL COMMENTS
DATE: DATE:

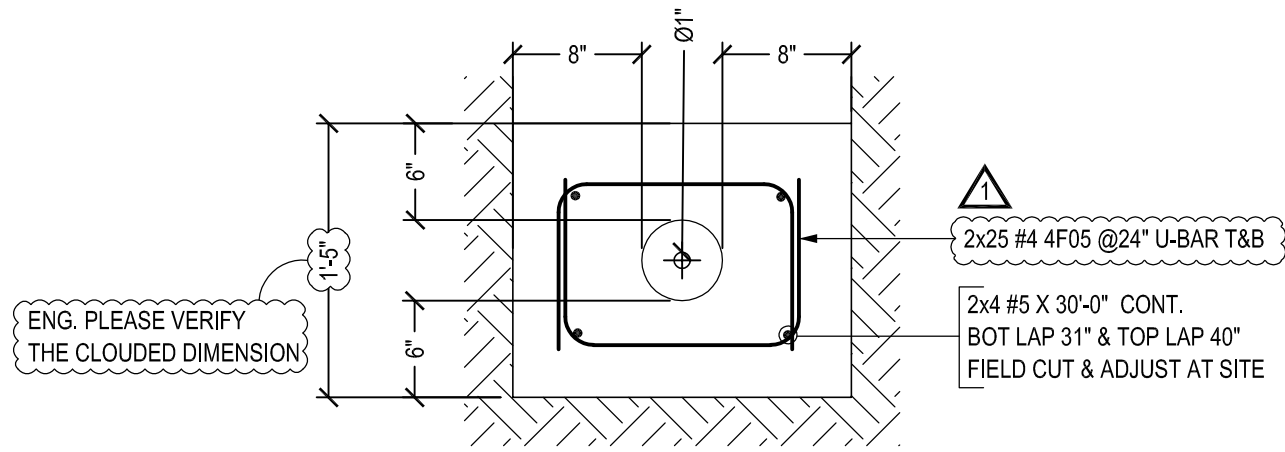
JOB #: 2524240001
DETAILS: MCR
EMAIL: will@cmcrebar.com
PHONE: (360)-346-6729
REF: 10-M-01-101

PROJECT INFORMATION
CUSTOMER: -
JOB NAME: HLO WWP
LOCATION: COUNTY OF HAWAII
DESCRIPTION: SLUDGE BLENDING BUILDING - PIPE ENCASEMENT REINFORCING



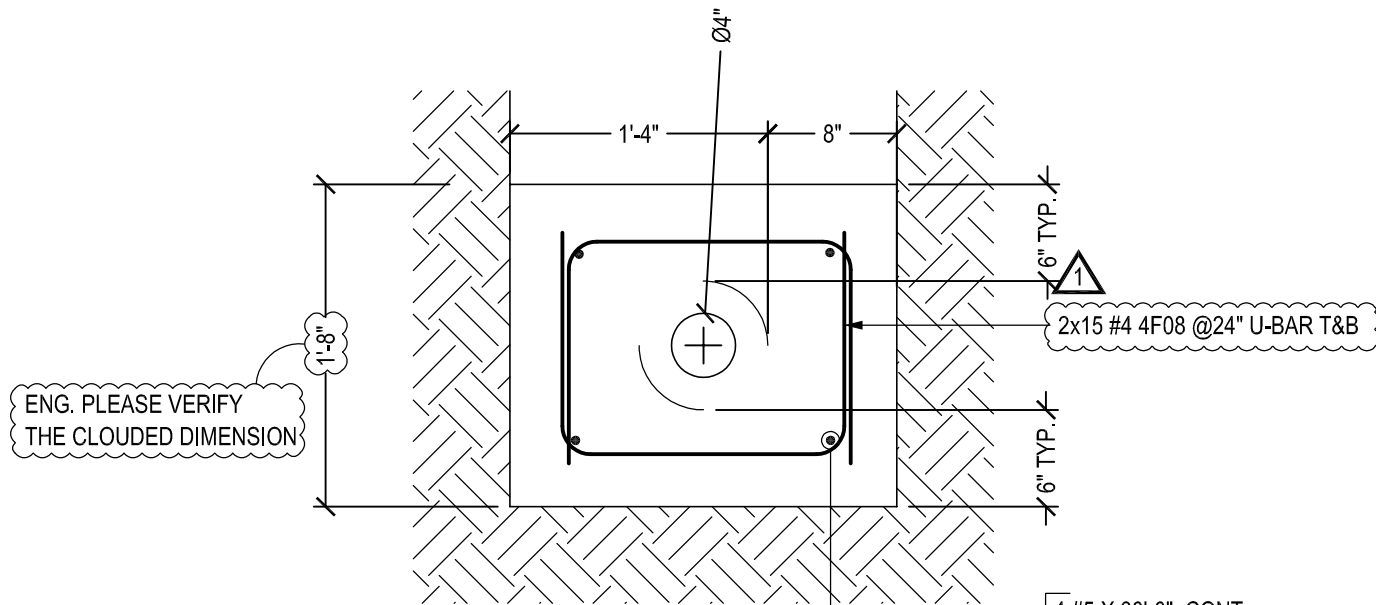
JOB#: 2524240001
DRAWING: RP-009

ALL REINFORCEMENTS ARE A615 GR.60 U.N.O



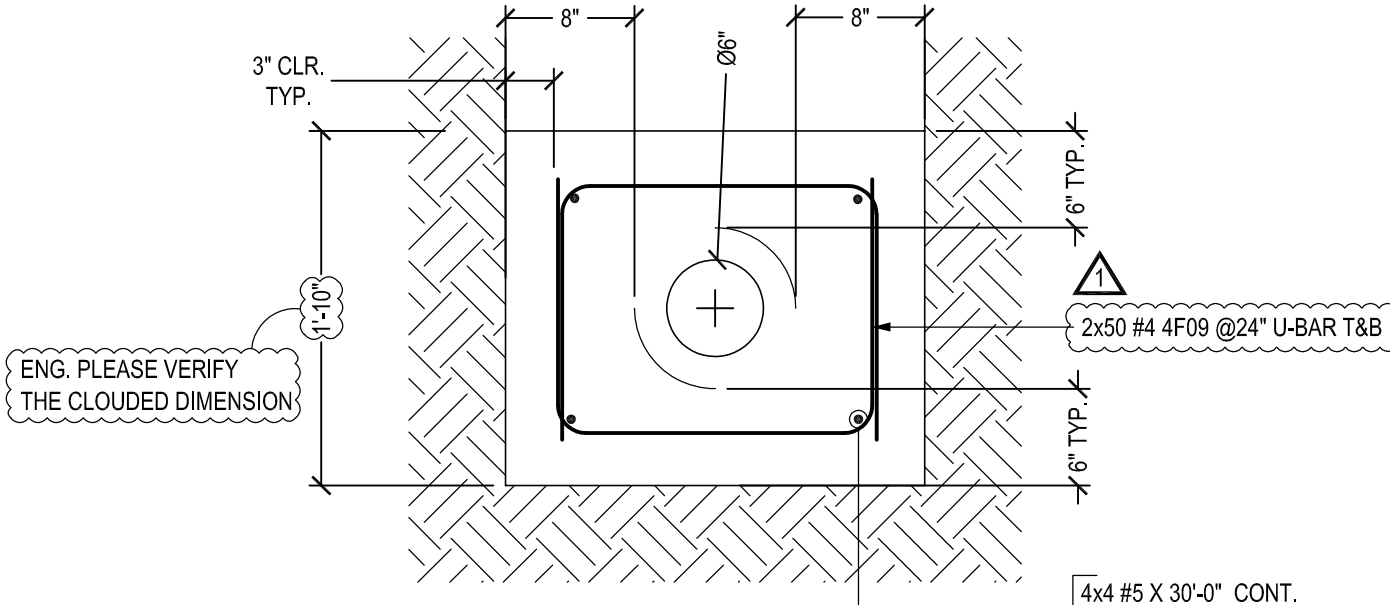
TYPICAL 1"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/11"=1'-0"
REF:00-T-05-702



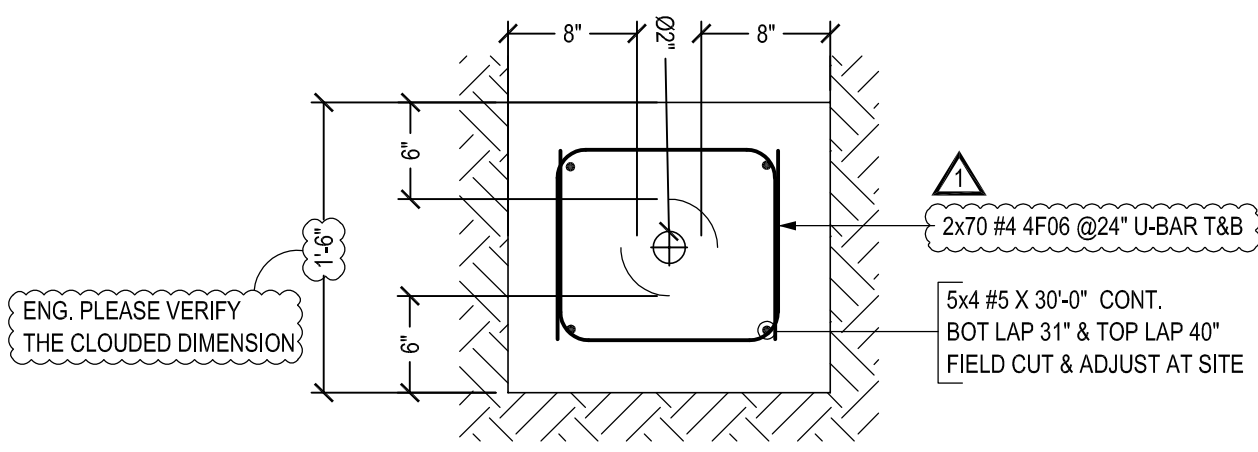
TYPICAL 4"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/11"=1'-0"
REF:00-T-05-702



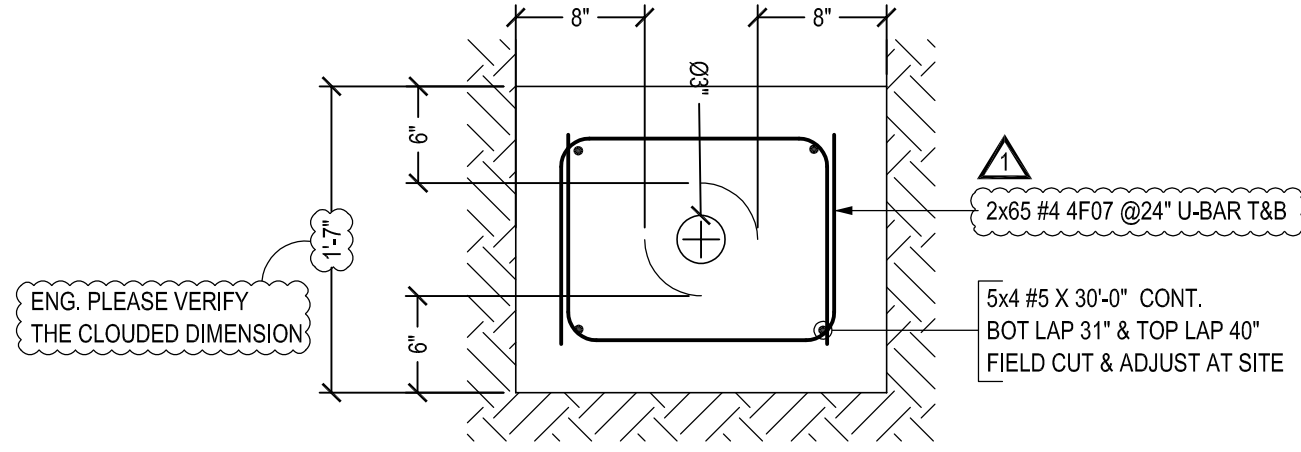
TYPICAL 6"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 2"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 3"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/11"=1'-0"
REF:00-T-05-702

Release Number: RP-009A-RP-010										BAR LIST									
Bar Mark	Qty	Size	Total Length	Type	X	Y	Z	W	V	U	T	S	R	Q	P	O	N	M	L
5F03	68	#5	5'-2"	17				2'-3"	2'-3"										
4F01	400	#4	4'-8"	17				1'-6"	1'-6"										
5F02	68	#5	3'-5"	2	0'-10"			2'-3"											
4F02	100	#4	4'-8"	17				1'-6"	1'-6"										
5F01	36	#5	10'-10"	2	0'-10"			10'-0"											
4F04	100	#4	4'-0"	17				1'-4"	1'-4"	1'-4"									
4F03	80	#4	3'-9"	17				1'-3"	1'-3"	1'-3"									
4F09	100	#4	4'-4"	17				1'-4"	1'-4"	1'-4"									
4F08	30	#4	3'-10"	17				1'-2"	1'-2"	1'-2"									
4F07	130	#4	3'-7"	17				1'-1"	1'-5"	1'-1"									
4F06	140	#4	3'-2"	17				1'-0"	1'-2"	1'-0"									
4F05	50	#4	3'-0"	17				0'-10 1/2"	1'-3"	0'-10 1/2"									

For list produced by RobotCAD 2000

17	02
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ABBREVIATIONS	
VERTICAL	VERT.
EACH FACE	EF
INNER FACE	IF
OUTER FACE	OF
EACH SIDE	ES
DOWEL	DWL
REFERENCE	REF.

LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#3	24"	19"
#4	32"	25"
#5	40"	31"
#6	48"	37"
#7	70"	54"
#8	80"	62"
#9	91"	70"
#10	102"	79"

ENG. PLEASE VERIFY THE LAP SCHEDULE

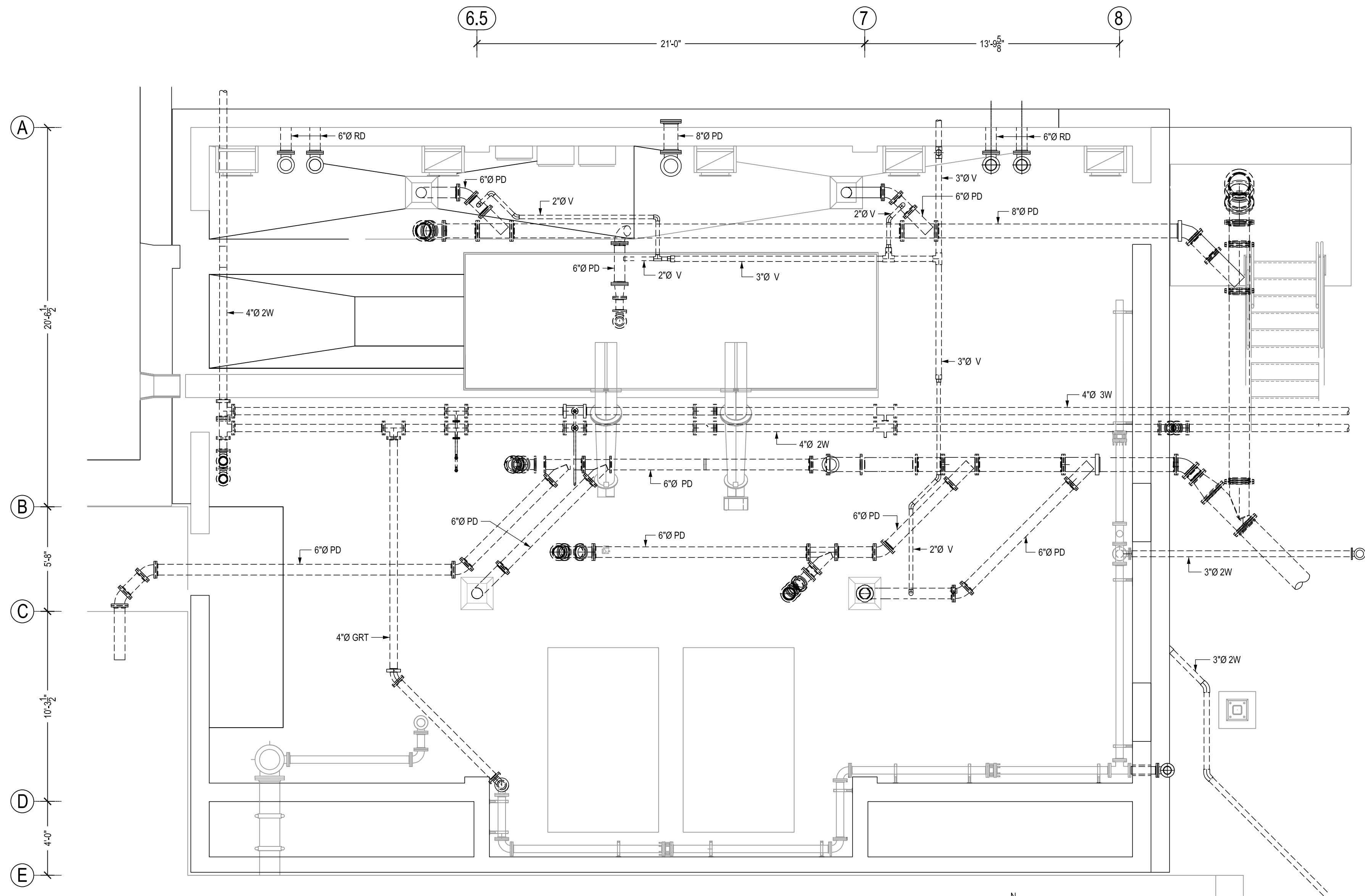
FIELD USE:
THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND INSTALLING REINFORCEMENT. THE CONTRACTOR SHALL BE RESPONSIBLE FOR LAYOUT AND CONSTRUCTION. ANY DIMENSIONS OR NOTES ON THIS DRAWING SHALL BE USED TO FACILITATE CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING THE DIMENSIONS AND NOTES ON THIS DRAWING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING THE DIMENSIONS AND NOTES ON THIS DRAWING. THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING THE DIMENSIONS AND NOTES ON THIS DRAWING.

APPROVAL AUTHORITY:
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SUBMITTAL RECORD	
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DATE: .	FOR: .
DATE: 06/13/25	REVISD AS PER MAIL COMMENTS
DATE: .	DATE: .

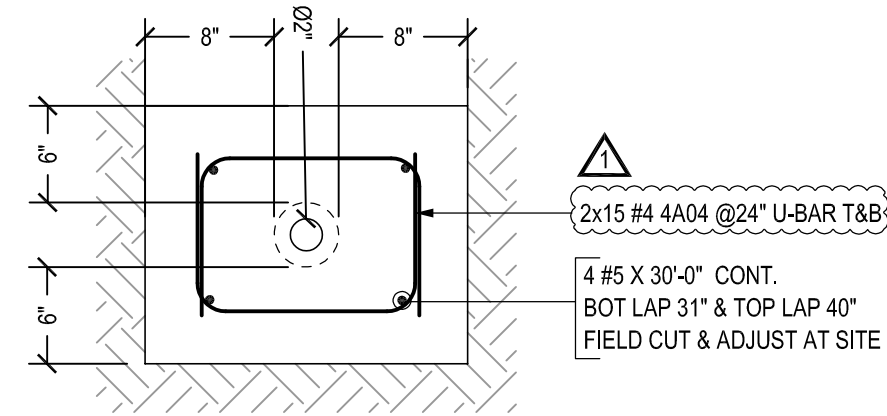
PROJECT INFORMATION	
CUSTOMER: -	JOB #: 2524240001
JOB NAME: HLO WWP	DETAILS: MQR
LOCATION: COUNTY OF HAWAII	EMAIL: willow.construction.com
DESCRIPTION: SLUDGE BLENDING BUILDING - PIPE ENCASEMENT REINFORCING	PHONE: (360) 346-6729
	REF: 10-M-01-101





SCREENINGS & GRIT HANDLING - PIPE ENCASEMENT LAYOUT

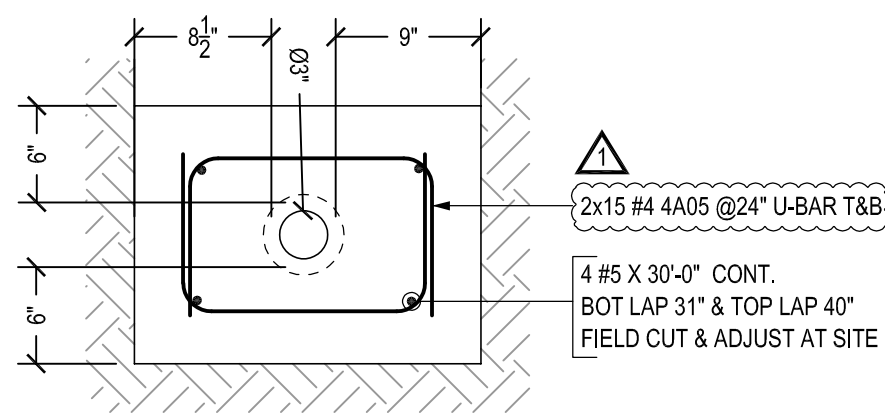
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TYPICAL 2" PIPE ENCASEMENT REINFORCEMENT DETAILS

TOTAL LENGTH = 25'-3"

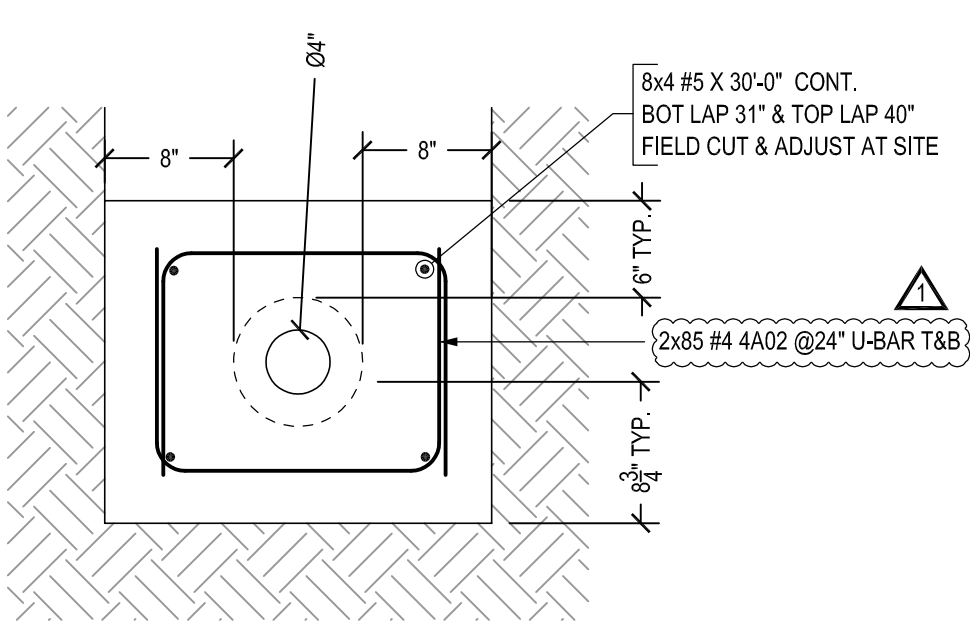
SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 3" PIPE ENCASEMENT REINFORCEMENT DETAILS

TOTAL LENGTH = 29'-0"

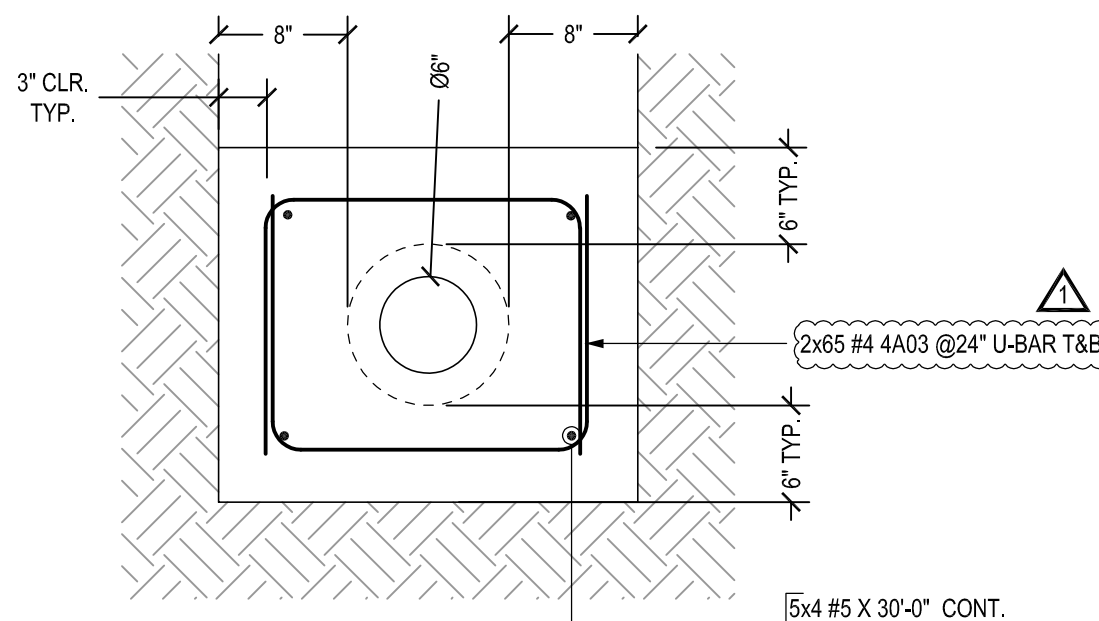
SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 4" PIPE ENCASEMENT REINFORCEMENT DETAILS

TOTAL LENGTH = 167'-6"

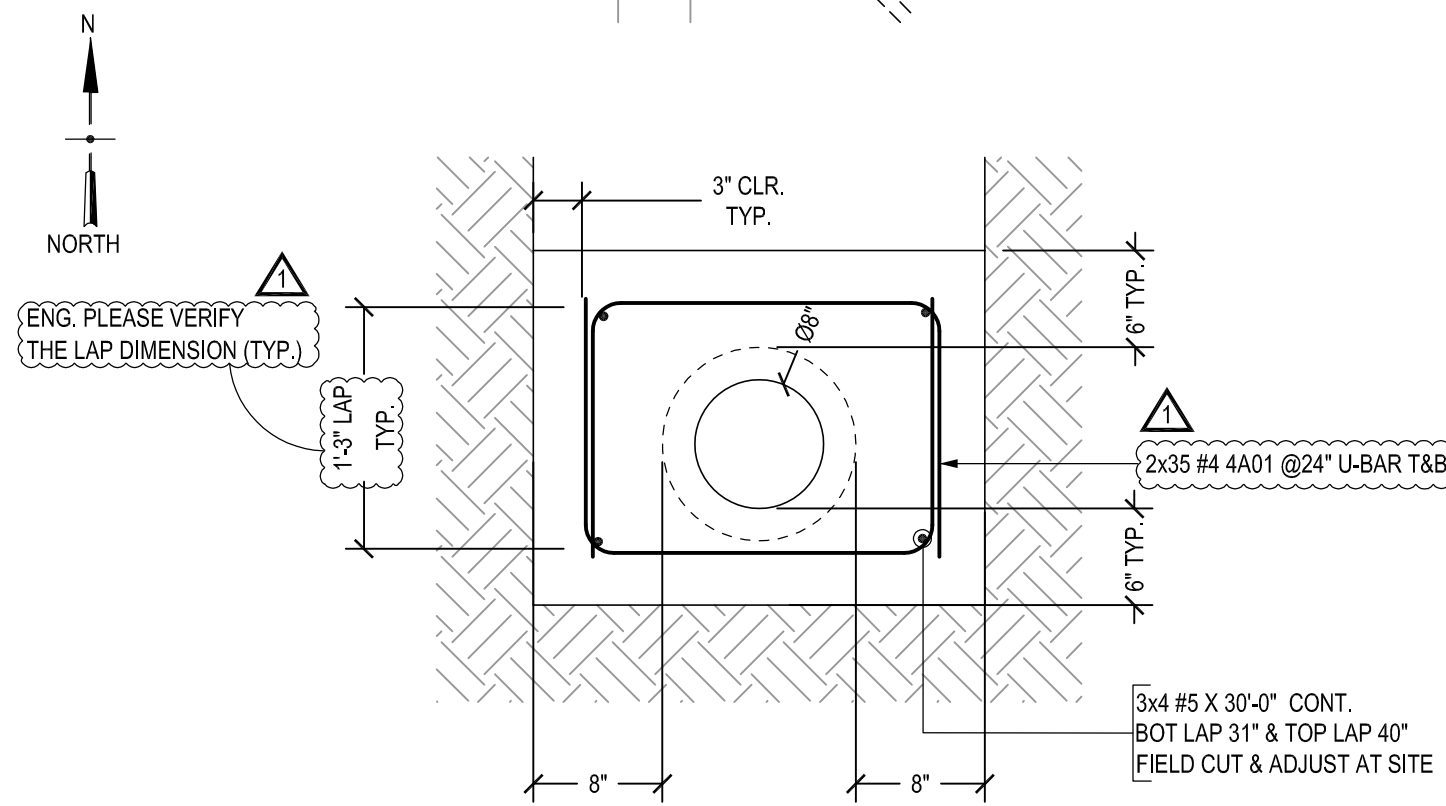
SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 6" PIPE ENCASEMENT REINFORCEMENT DETAILS

TOTAL LENGTH = 129'-6"

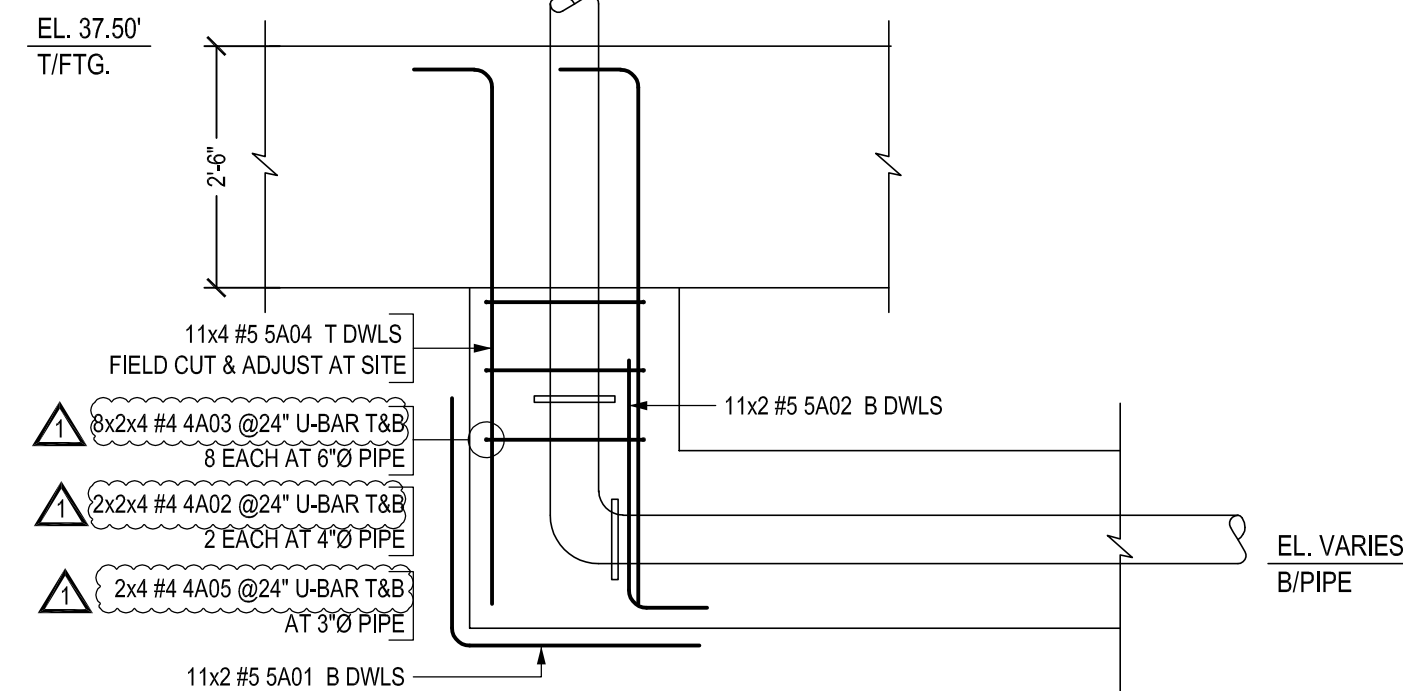
SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL 8" PIPE ENCASEMENT REINFORCEMENT DETAILS

TOTAL LENGTH = 69'-6"

SCALE - 1/11"=1'-0"
REF:00-T-05-702



TYPICAL PIPE ENCASEMENT UNDER SLAB CONNECTION DETAILS

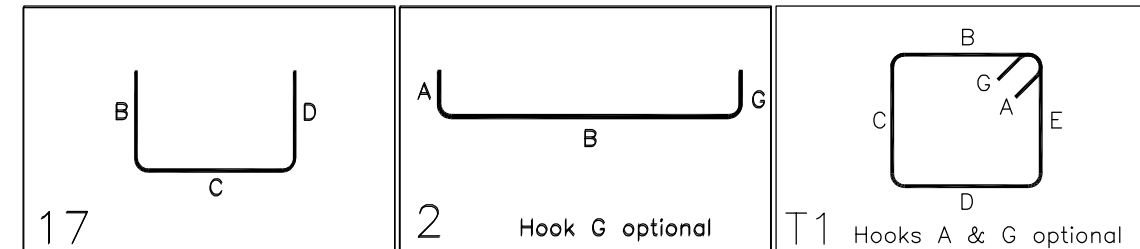
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SCALE - 1/12"=1'-0"
REF:00-T-05-702

Release Number: RP-02				BAR LIST													
Bar Mark	Qty	Size	Total Length	Type	'A'	'B'	'C'	'D'	'E'	'F'	'G'	'H'	'J'	'K'	'L'	'M'	
4401	70	#4	4'-6"	17			1'-4"	1'-10"	1'-4"								
4402	186	#4	3'-10"	17			1'-2"	1'-6"	1'-2"								
4403	194	#4	4'-4"	17			1'-4"	1'-8"	1'-4"								
5402	22	#5	3'-5"	2	0'-10"		2'-7"										
5401	22	#5	5'-2"	17			2'-7"	2'-7"									
5404	44	#5	10'-10"	2	0'-10"		10'-0"										
4404	30	#4	2'-10"	17			0'-10"	1'-2"	0'-10"								
4405	38	#4	2'-11 3/4"	17			0'-10"	1'-3 3/4"	0'-10"								

Bar list produced by RebarCAD 2020.

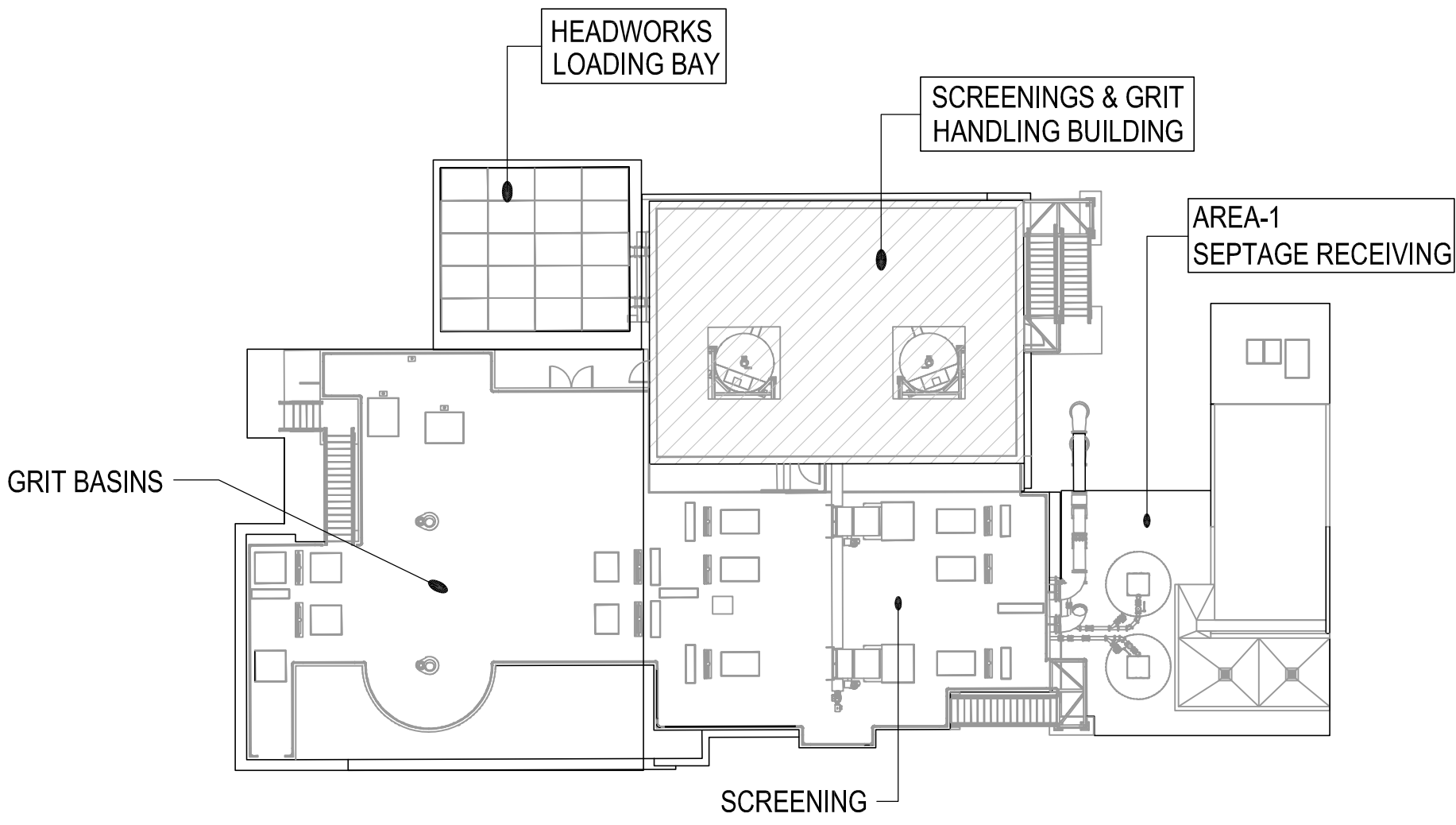
Bar list produced by RebarCAD 2020.0



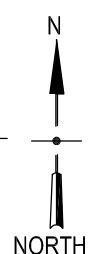
ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
CORNER BAR	COR.
REFERENCE	REF.

LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#3	24"	19"
#4	32"	25"
#5	40"	31"
#6	48"	37"
#7	70"	54"
#8	80"	62"
#9	91"	70"
#10	102"	79"

(ENG. PLEASE VERIFY THE LAP SCHEDULE)



KEY PLAN
SCALE - N.T.S
REF:02-M-01-105



ALL REINFORCEMENTS ARE A615 GR.60 U.N.O

PROJECT INFORMATION

CUSTOMER: -

JOB NAME: HLO WWP

LOCATION: COUNTY OF HAWAII

DESCRIPTION: SCREENINGS & GRIT HANDLING - PIPE ENCASEMENT DETAILS

JOB # 2524240001

DETAILS: MCR

EMAIL: will@cmcrebar.com

PHONE: (380)-346-6729

CNC

CNC REDAR

APPROVAL AUTHORITY:

ALL ITEMS THAT ARE CIRCLED OR MARKED "VERIFY" REFER TO QUESTIONS, OMISSIONS, CONFLICTS OR ERRORS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. SO MARKED WILL BE ASSUMED CORRECT. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS.

DATE: 06/13/25 FOR: APPROVAL

DATE: . FOR: .

DATE: 06/13/25 - REVISED AS PER MAIL COMMENTS

DATE: .

FIELD USE:

THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND CONCRETE. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS. THE USER SHALL BE RESPONSIBLE FOR VERIFYING ALL ITEMS.

DATE: 06/13/25 FOR: APPROVAL

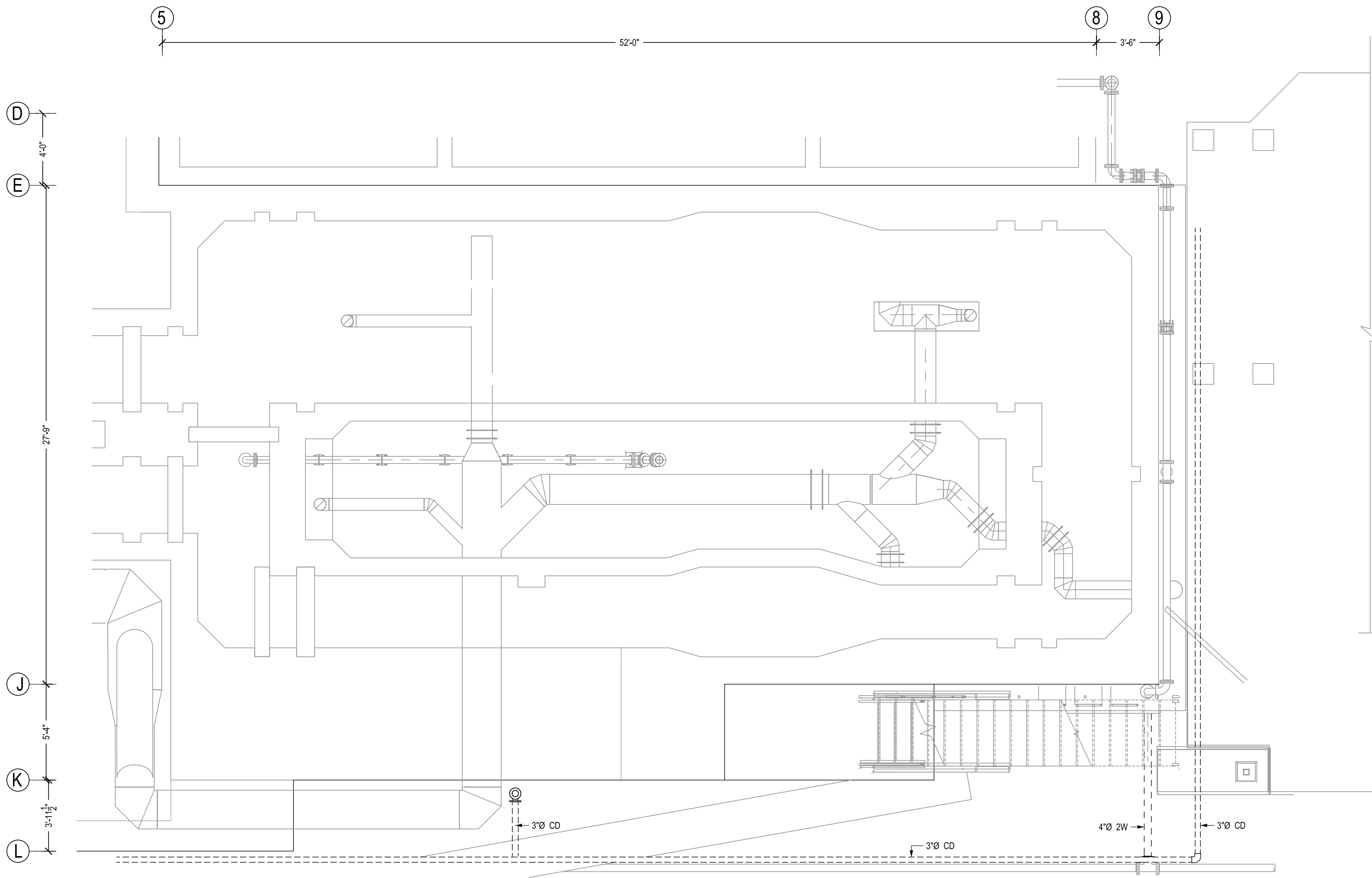
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DATE: 06/13/25 - REVISED AS PER MAIL COMMENTS

DATE: .

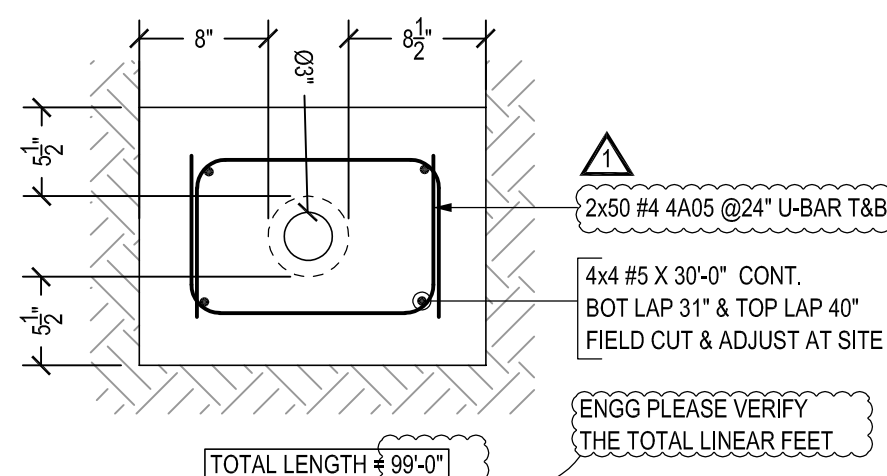
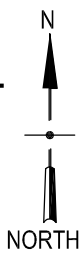
JOB# 2524240001

DRAWING: RP-02



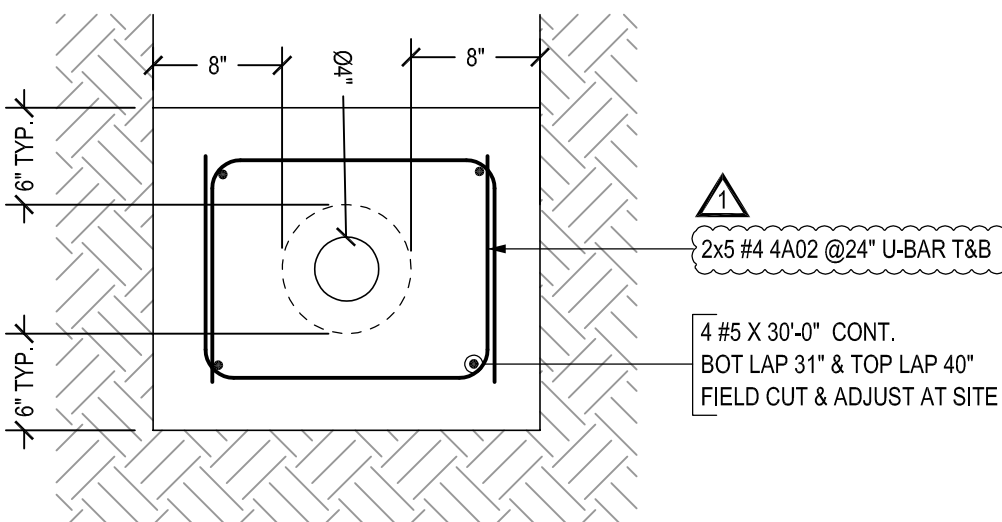
SCREENING CHANNEL - PIPE ENCASEMENT LAYOUT

SCALE - 1/4"=1'-0"
REF:02-M-01-103



TYPICAL 3"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/1"=1'-0"
REF:00-T-05-702

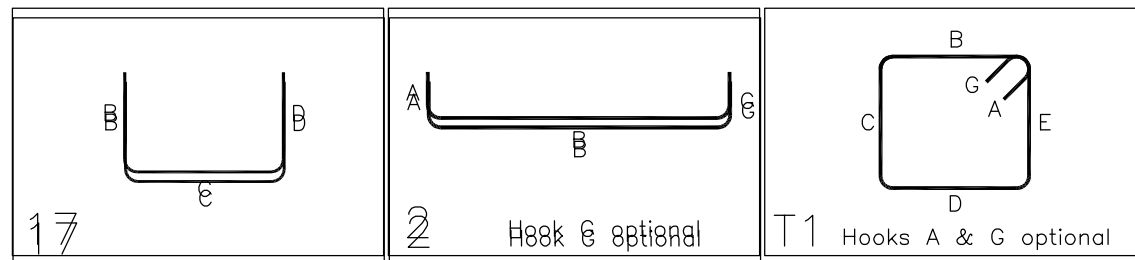


TYPICAL 4"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/1"=1'-0"
REF:00-T-05-702

BAR LIST													
Bar Mark	Qty	Size	Total Length	Type	'A'	'B'	'C'	'D'	'E'	'F'	'G'	'H'	'I'
5A01	2	#5	5'-2"	17		2'-7"	2'-7"						
5A02	2	#5	3'-5"	2	0'-10"	2'-7"							
4A05	106	#4	2'-11 1/8"	17		0'-10"	1'-3 3/8"	0'-10"					
5A04	4	#5	10'-10"	2	0'-10"	10'-0"							
4A02	10	#4	3'-10"	17		1'-2"	1'-6"	1'-2"					

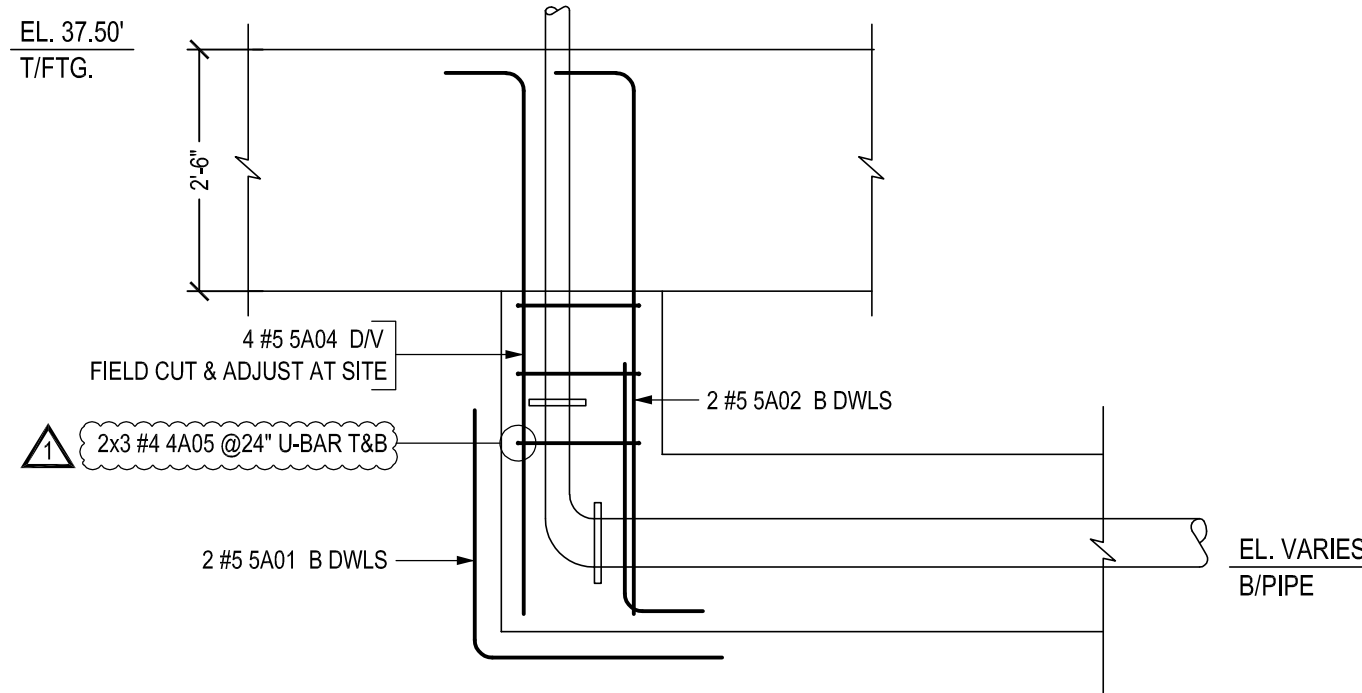
Bar list produced by RebarCAD 2020.0



ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
CORNER BAR	COR.
REFERENCE	REF.

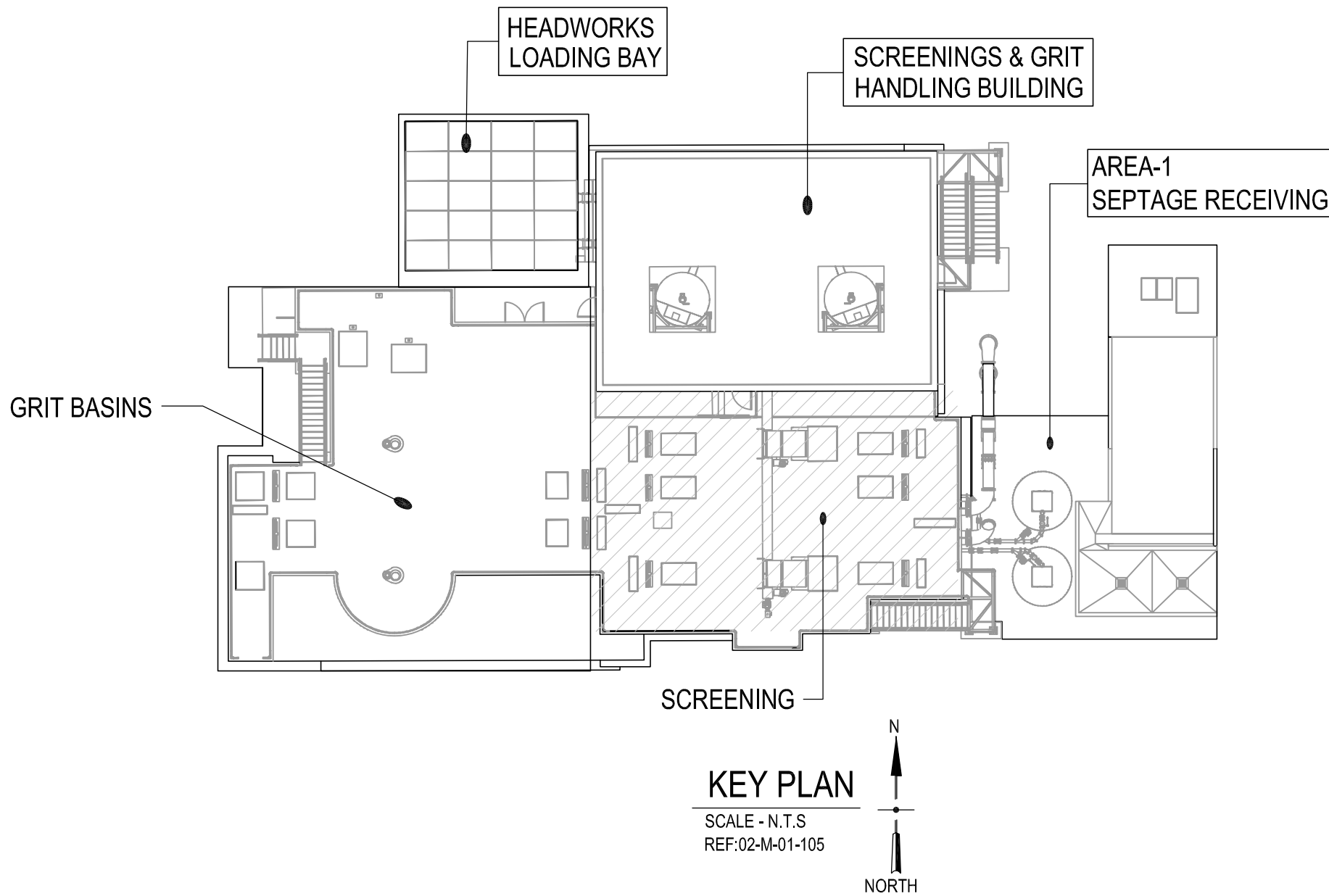
LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#3	24"	19"
#4	32"	25"
#5	40"	31"
#6	48"	37"
#7	70"	54"
#8	80"	62"
#9	91"	70"
#10	102"	79"

ENG. PLEASE VERIFY
THE LAP SCHEDULE



TYPICAL 3"Ø PIPE ENCASEMENT UNDER SLAB CONNECTION DETAILS

SCALE - 1/2"=1'-0"
REF:00-T-05-702



KEY PLAN

SCALE - N.T.S.
REF:02-M-01-105

ALL REINFORCEMENTS ARE A615 GR.60 U.N.O.

FIELD USE:
THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE REBAR CAD SOFTWARE AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND CONCRETE CASTING. IT IS NOT TO BE USED FOR ANY OTHER PURPOSES. FOR LAYOUT OR CONSTRUCTION, ANY DIMENSIONS OR NOTES ON THIS DRAWING SHALL TAKE PRECEDENCE OVER ANY DIMENSIONS OR NOTES ON ANY OTHER DRAWING. APPROVAL OF THIS DRAWING IS THE RESPONSIBILITY OF THE ENGINEER. APPROVAL OF THIS DRAWING IS THE RESPONSIBILITY OF THE ENGINEER. APPROVAL OF THIS DRAWING IS THE RESPONSIBILITY OF THE ENGINEER.

APPROVAL AUTHORITY:
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SUBMITTAL RECORD	
DATE: 06/13/25	FOR: APPROVAL
DATE: .	FOR: .
DATE: .	FOR: .
DATE: 06/13/25	REVISION: AS PER MAIL COMMENTS
DATE: .	REVISION: .
DATE: .	REVISION: .

PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED

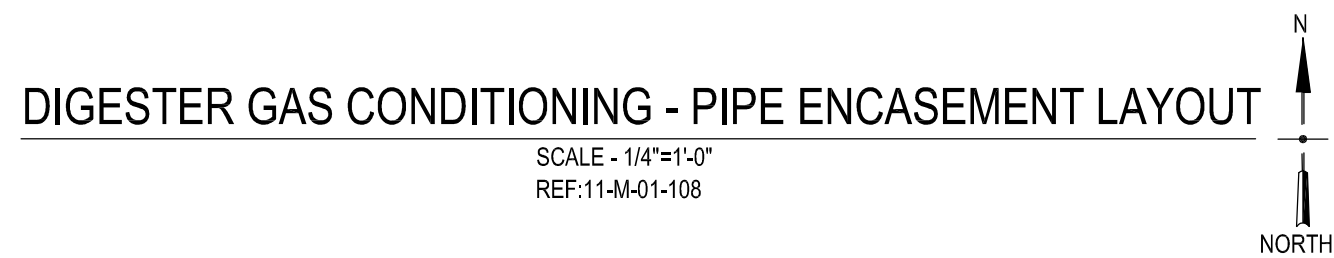
PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED

PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED

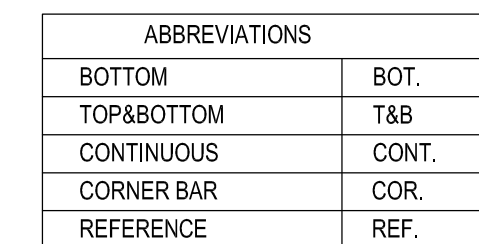
PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED

PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED

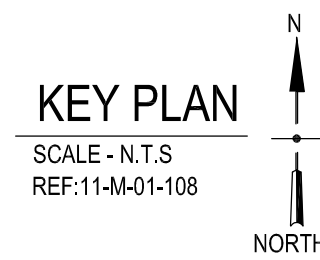
PROJECT INFORMATION	
JOB #: 2524240001	DETAILS: MCR
EMAIL: will@concretebar.com	PHONE: (360) 346-0729
CUSTOMER: HLO WWP	LOCATION: COUNTY OF HAWAII
DESCRIPTION: SCREENINGS CHANNEL - PIPE ENCASEMENT DETAILS	REF: NOTED



Bar list produced by RebarCAD 2020.



ENG. PLEASE VERIFY
THE LAP SCHEDULE



THIS DRAWING TO BE USED IN CONJUNCTION w/ STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING OF REINFORCING BARS ONLY. DO NOT USE THIS DRAWING FOR LAYOUT OR CONSTRUCTION. ANY DIMENSIONS OR ELEVATIONS NOTED IN THIS DOCUMENT ARE FOR REFERENCE ONLY. TO FACILITATE APPROVAL &/OR PLACEMENT OF REBAR, DISCREPANCIES SHOULD BE NOTED AT THE TIME OF APPROVAL. CM/C REBAR ASSUMES NO LIABILITY FOR DIMENSIONAL ERRORS IN THE LAYOUT OF THE BUILDING OR FORMS.

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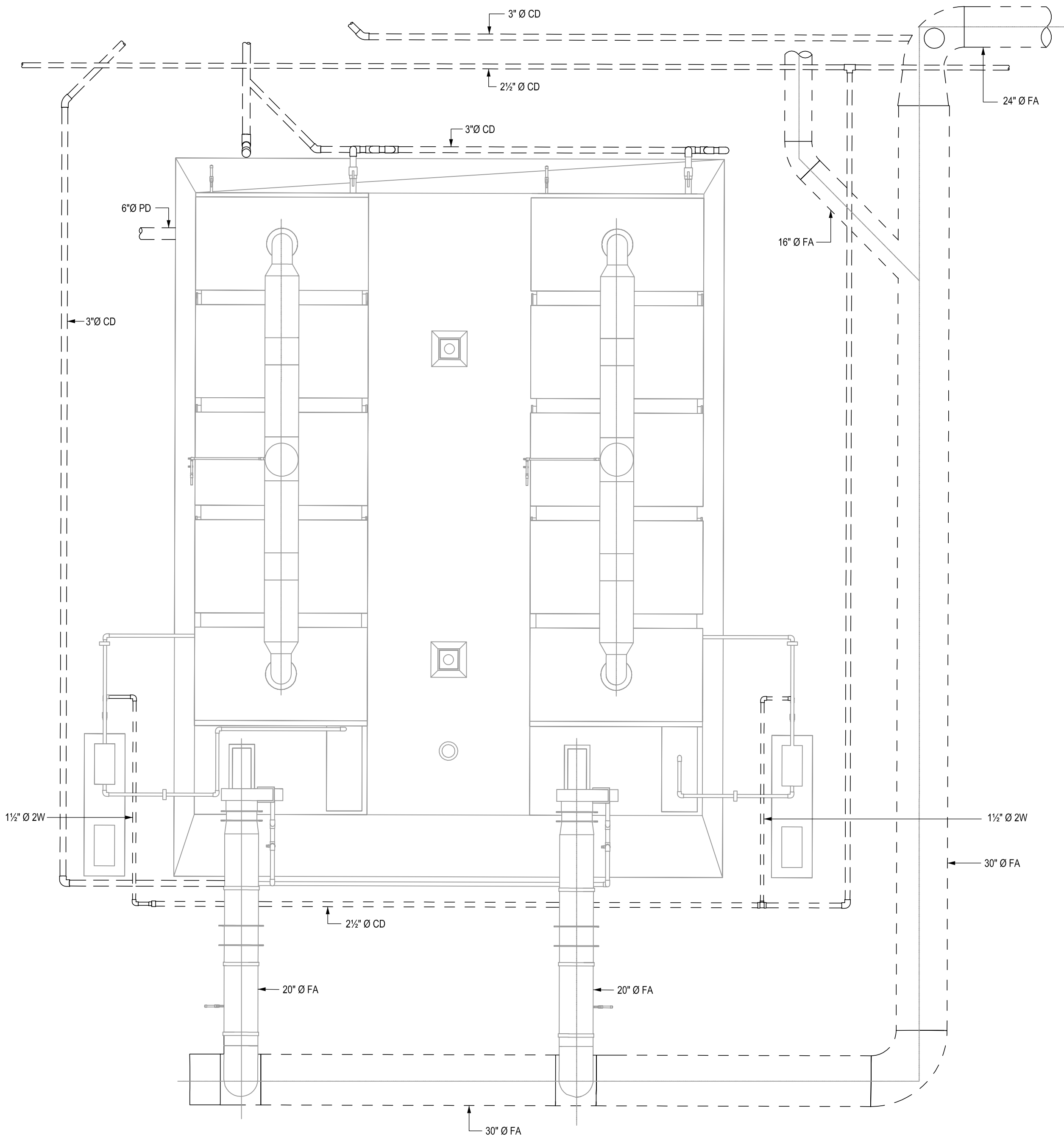
DATE	DATE
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PHONE: (360)-346-6729

COUNTY OF HAWAII

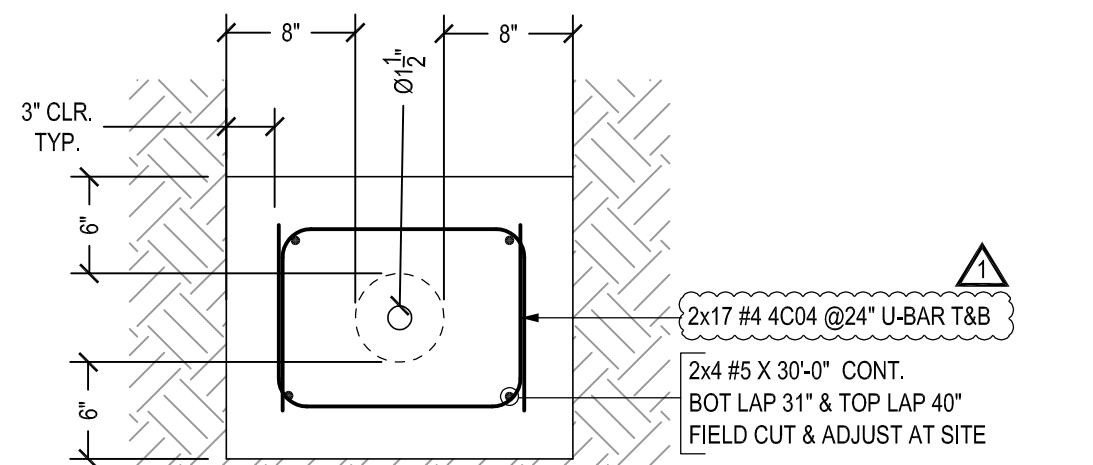
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ALL REINFORCEMENTS ARE A615 GR.60 U.N.O



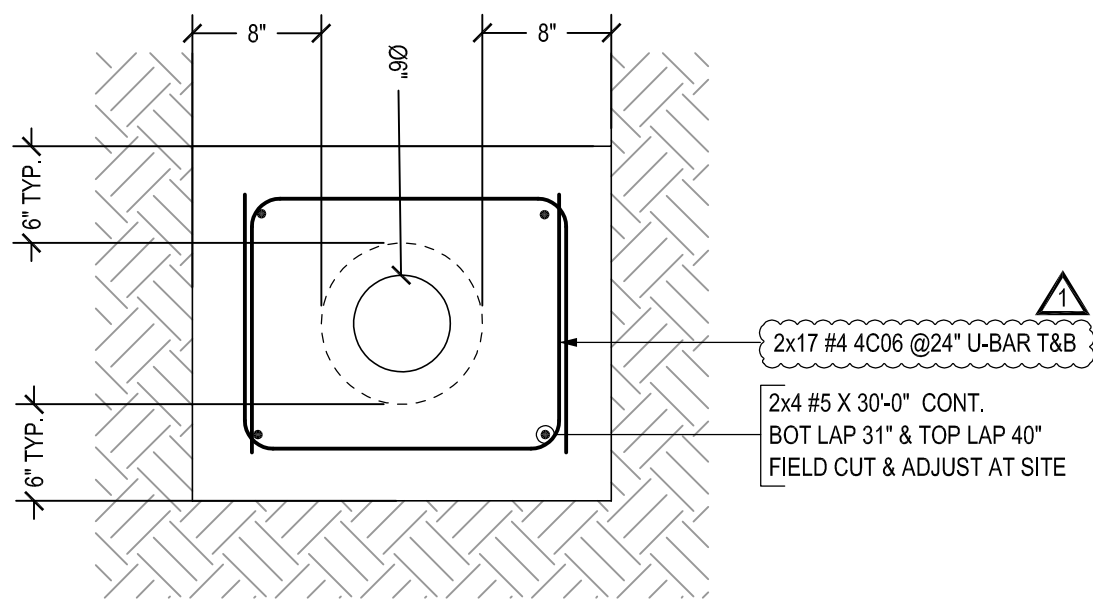
SOLIDS ODOUR CONTROL BUILDING - PIPE ENCASEMENT LAYOUT

SCALE - 1/8"=1'-0"
REF:14-M-01-101



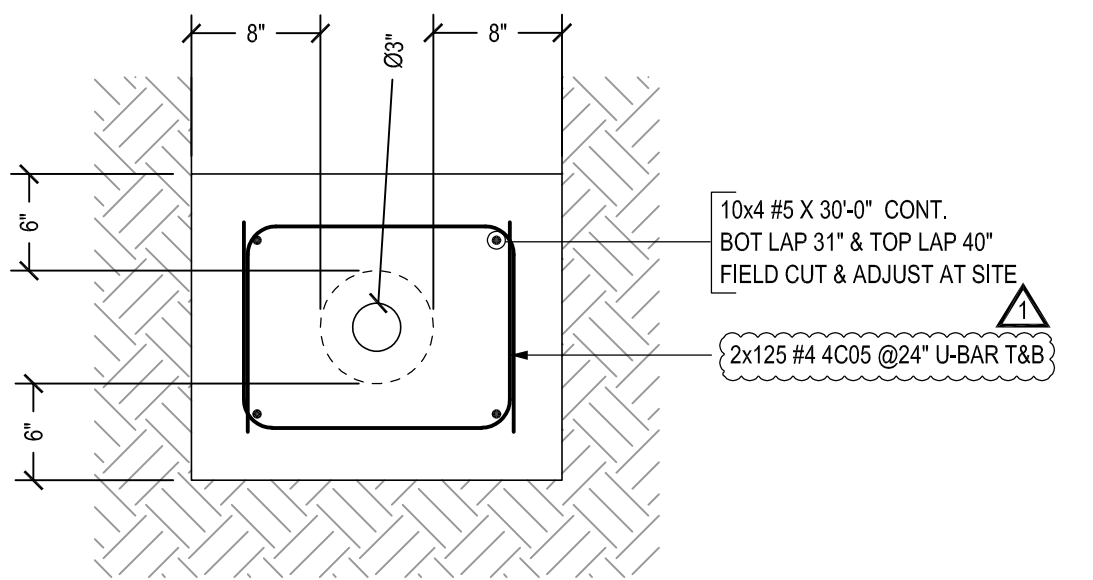
TYPICAL 1 1/2"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/1"=1'-0"
REF:00-T-05-702



TYPICAL 6"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/1"=1'-0"
REF:00-T-05-702

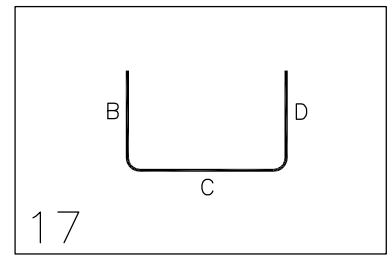


TYPICAL 2 1/2"Ø & 3"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

SCALE - 1/1"=1'-0"
REF:00-T-05-702

Release Number: RP-05					BAR LIST												
Bar Mark	Qty	Size	Total Length	Type	A'	B'	C'	D'	E'	F'	G'	H'	I'	J'	K'	L'	M'
4C04	34	#4	3'-2 3/4"	17		0'-11 3/4"	1'-3 3/4"	0'-11 3/4"									
4C05	250	#4	3'-2 3/4"	17		1'-1"	1'-5"	1'-1"									
4C06	34	#4	4'-4"	17		1'-4"	1'-8"	1'-4"									

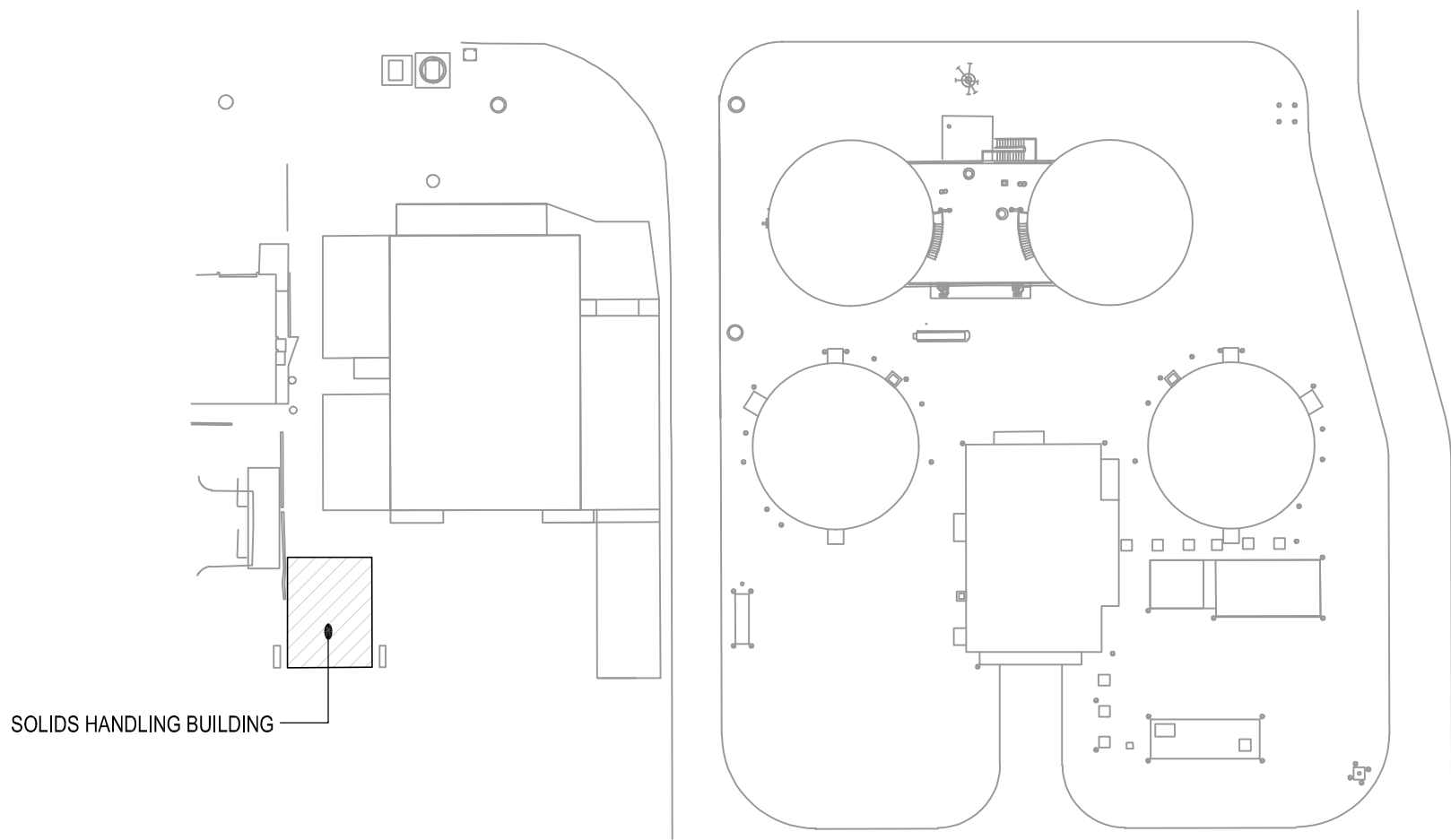
Bar list produced by RebarCAD 2020.0



ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
CORNER BAR	COR.
REFERENCE	REF.

LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#3	24"	19"
#4	32"	25"
#5	40"	31"
#6	48"	37"
#7	70"	54"
#8	80"	62"
#9	91"	70"
#10	102"	79"

ENG. PLEASE VERIFY
THE LAP SCHEDULE



KEY PLAN

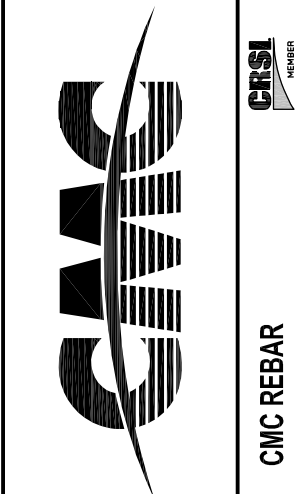
SCALE - N.T.S.
REF:11-M-01-108

SUBMITTAL RECORD

DATE	06/13/25	FOR	APPROVAL
DATE		FOR	
DATE		FOR	
DATE		FOR	

PROJECT INFORMATION

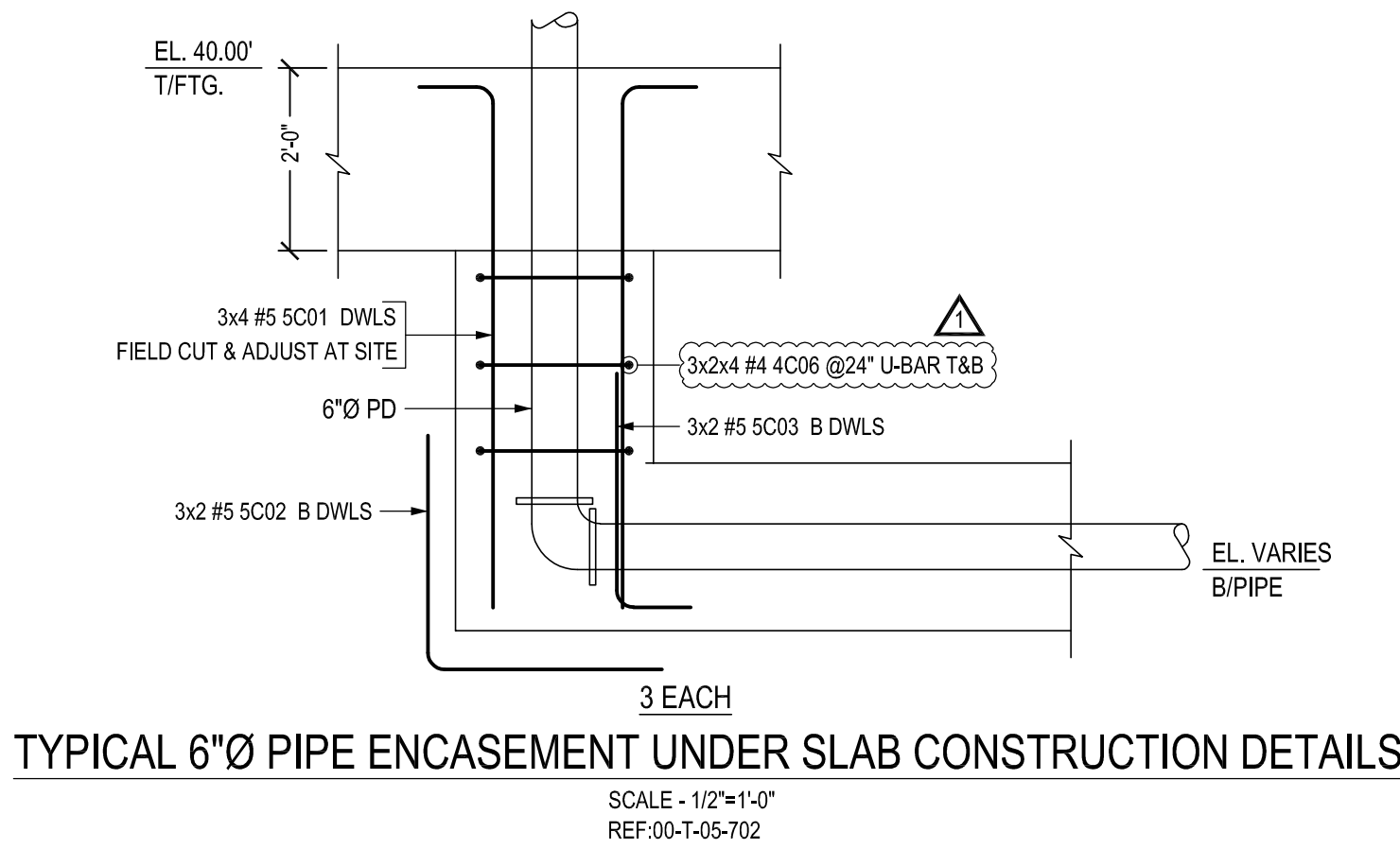
CUSTOMER	-	JOB #	2524240001
JOB NAME	HLO WWP	DETAILS	M/C
LOCATION	COUNTY OF HAWAII	EMAIL	will@cmcrebar.com
DESCRIPTION	SOLIDS ODOUR CONTROL BUILDING - PIPE ENCASEMENT DETAILS	PHONE	(808) 346-6729
		REF	NOTED



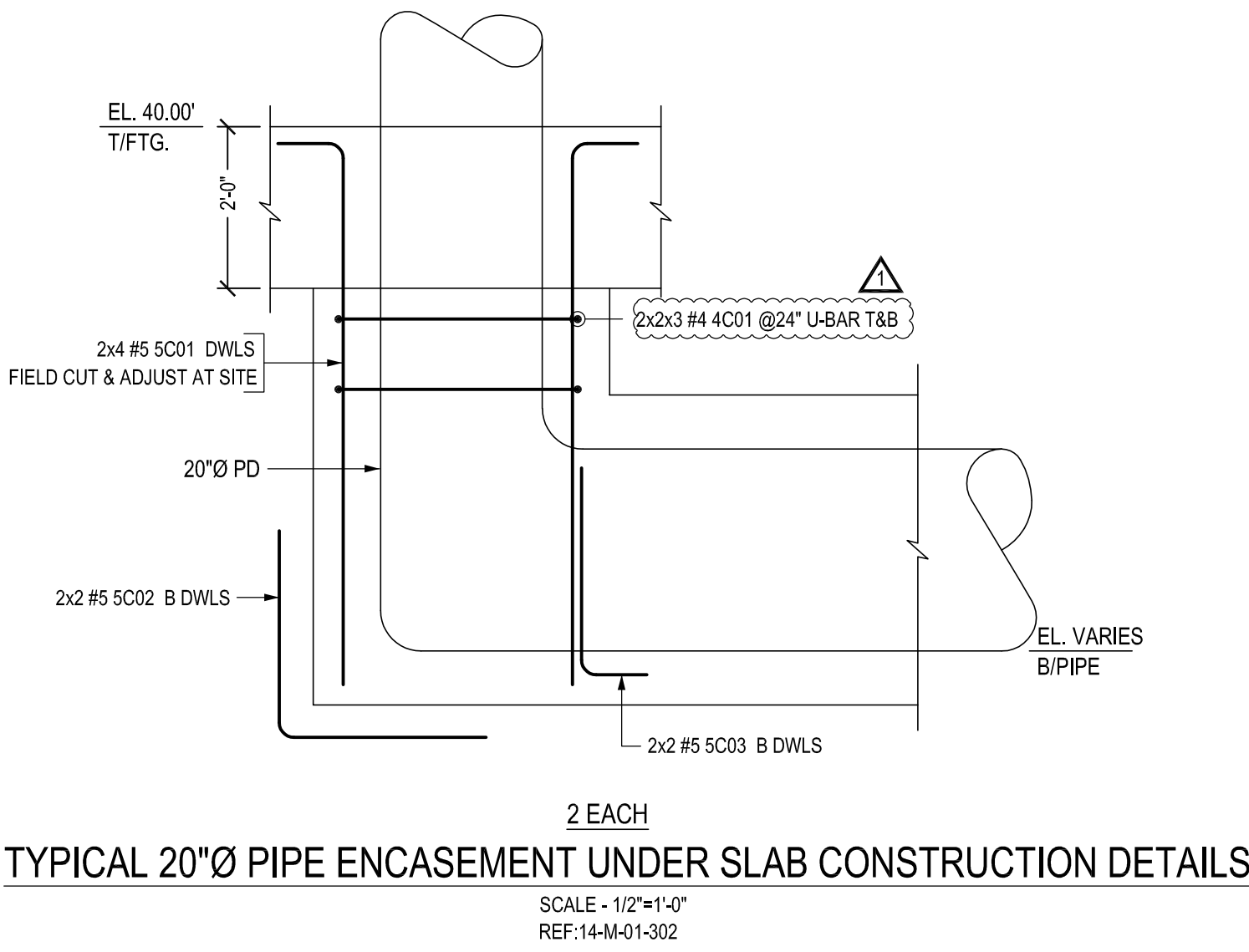
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DRAWING:
RP-05

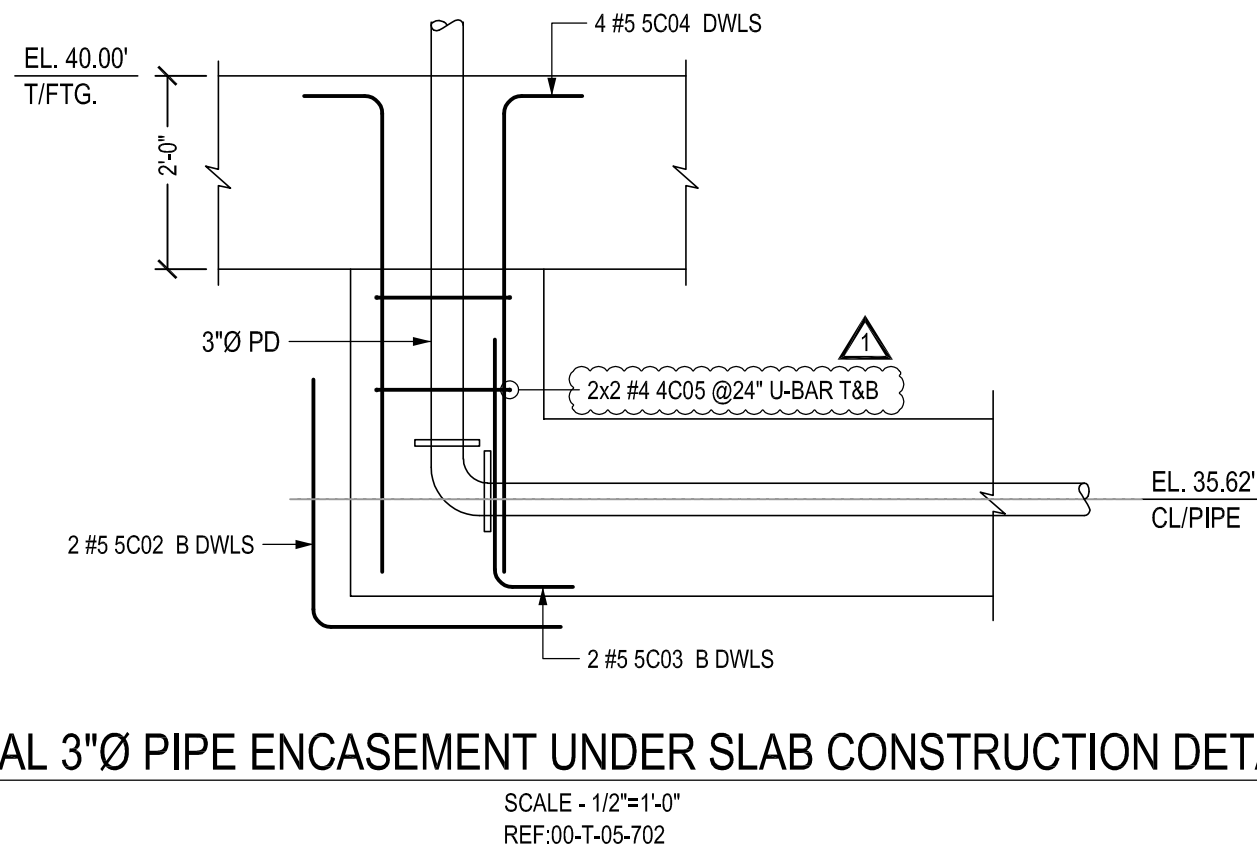
ALL REINFORCEMENTS ARE A706 GR.60 U.N.O



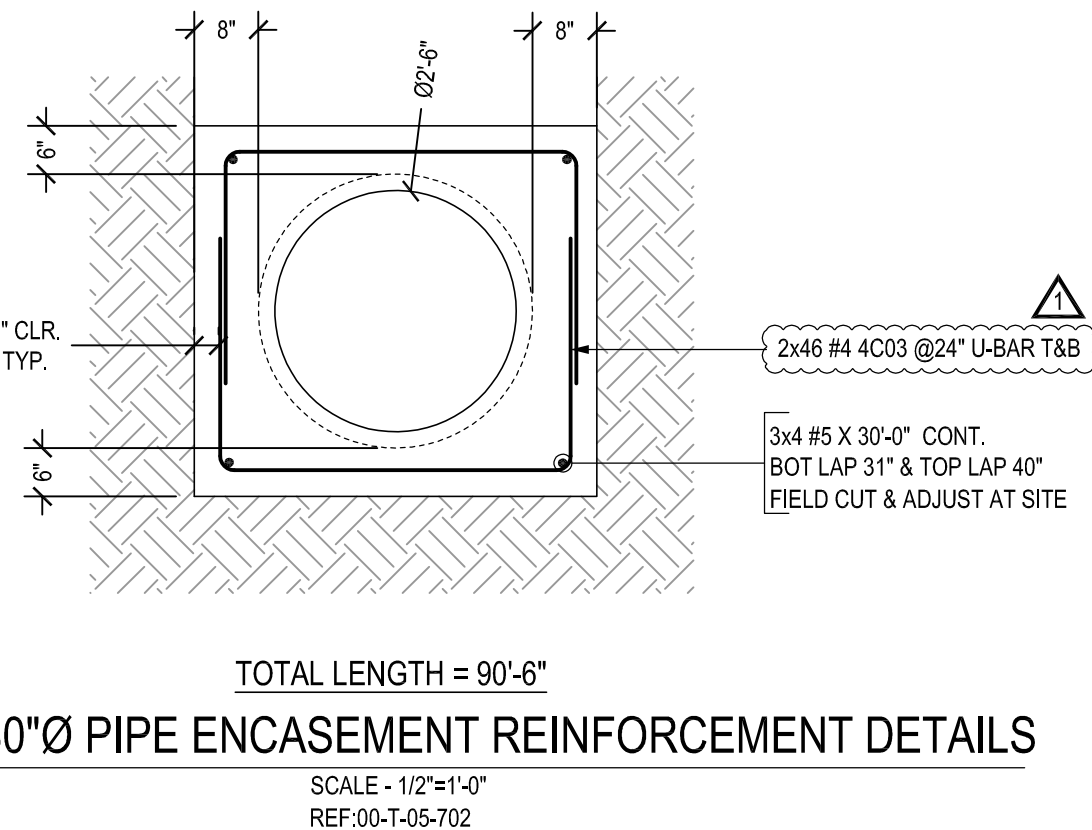
TYPICAL 6"Ø PIPE ENCASEMENT UNDER SLAB CONSTRUCTION DETAILS



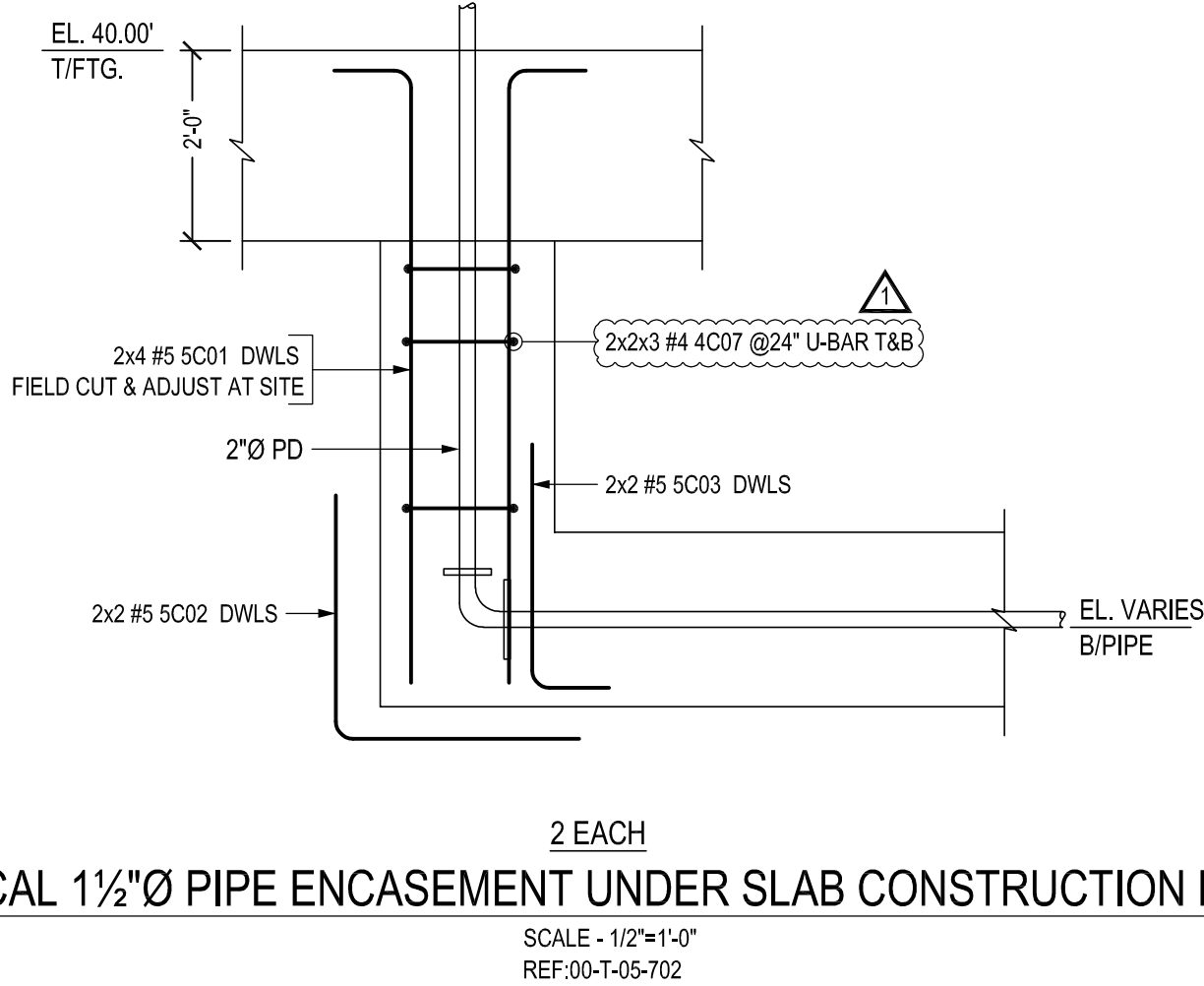
TYPICAL 20"Ø PIPE ENCASEMENT UNDER SLAB CONSTRUCTION DETAILS



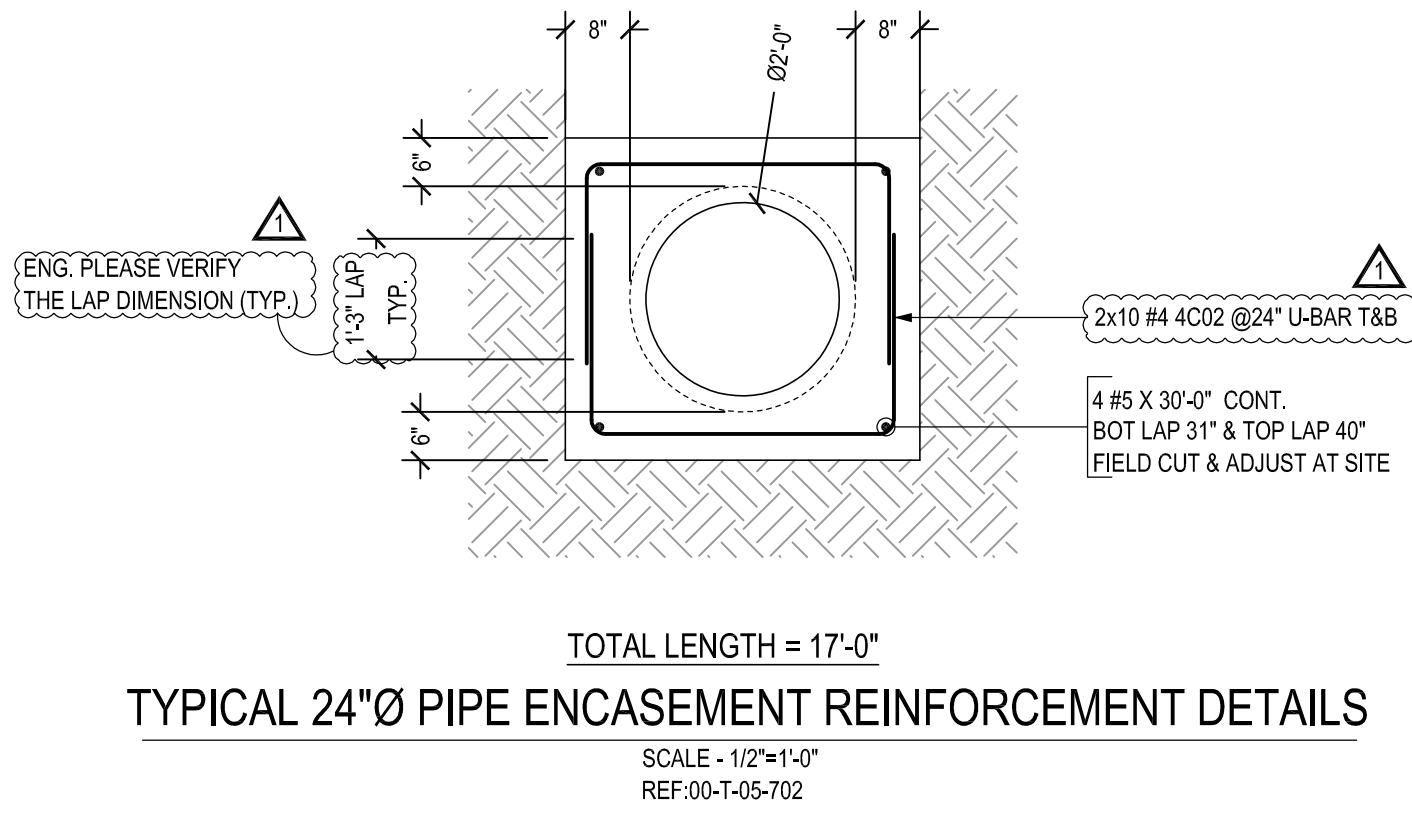
TYPICAL 3"Ø PIPE ENCASEMENT UNDER SLAB CONSTRUCTION DETAILS



TYPICAL 30"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS



TYPICAL 1½"Ø PIPE ENCASEMENT UNDER SLAB CONSTRUCTION DETAILS

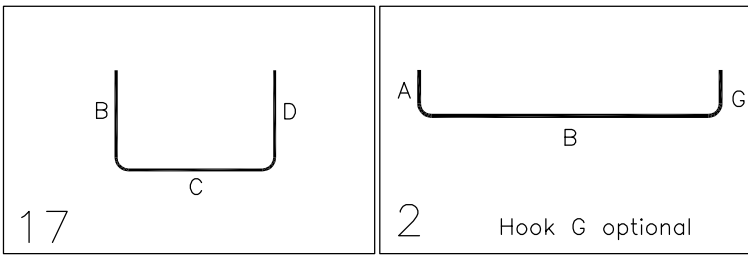


TYPICAL 24"Ø PIPE ENCASEMENT REINFORCEMENT DETAILS

Release Number: RP-06					BAR LIST												
Bar Mark	Qty	Size	Total Length	Type	'A'	'B'	'C'	'D'	'E'	'F'	'G'	'H'	'I'	'J'	'K'	'L'	'M'
4C01	12	#4	7'-0"	17		2'-0"	3'-0"	2'-0"									
4C02	20	#4	7'-4"	17		2'-1"	3'-2"	2'-1"									
4C03	92	#4	8'-6"	17		2'-5"	3'-8"	2'-5"									
4C05	4	#4	3'-7"	17		1'-1"	1'-5"	1'-1"									
4C06	24	#4	4'-4"	17		1'-4"	1'-8"	1'-4"									
4C07	12	#4	2'-10"	17		0'-10"	1'-2"	0'-10"									
5C01	28	#5	10'-10"	2	0'-10"	10'-0"											
5C02	16	#5	5'-2"	17			2'-7"										
5C03	16	#5	3'-5"	2	0'-10"	2'-7"											
5C04	4	#5	5'-10"	2	0'-10"	5'-0"											

Bar list produced by RebarCAD 2020.0

Bar list produced by RedarCAD 2020.0



ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
CORNER BAR	COR.
REFERENCE	REF.

LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#3	24"	19"
#4	32"	25"
#5	40"	31"
#6	48"	37"
#7	70"	54"
#8	80"	62"
#9	91"	70"
#10	102"	79"

ENG. PLEASE VERIFY
THE LAP SCHEDULE

FIELD USE:
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APPROVAL AUTHORITY:
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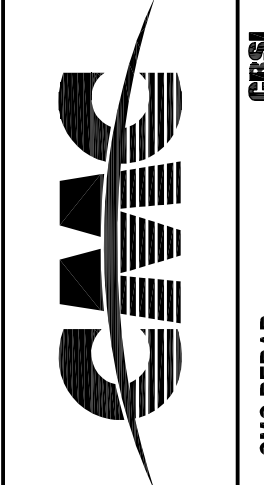
SUBMITTAL RECORD

DATE	06/13/25	FOR	APPROVAL
DATE		FOR	
DATE		FOR	

JOB #	2524240001
DETAILS	WIP
EMAIL	willem@concretebar.com
PHONE	(360) 346-6729
DATE	06/13/25
DATE	
DATE	

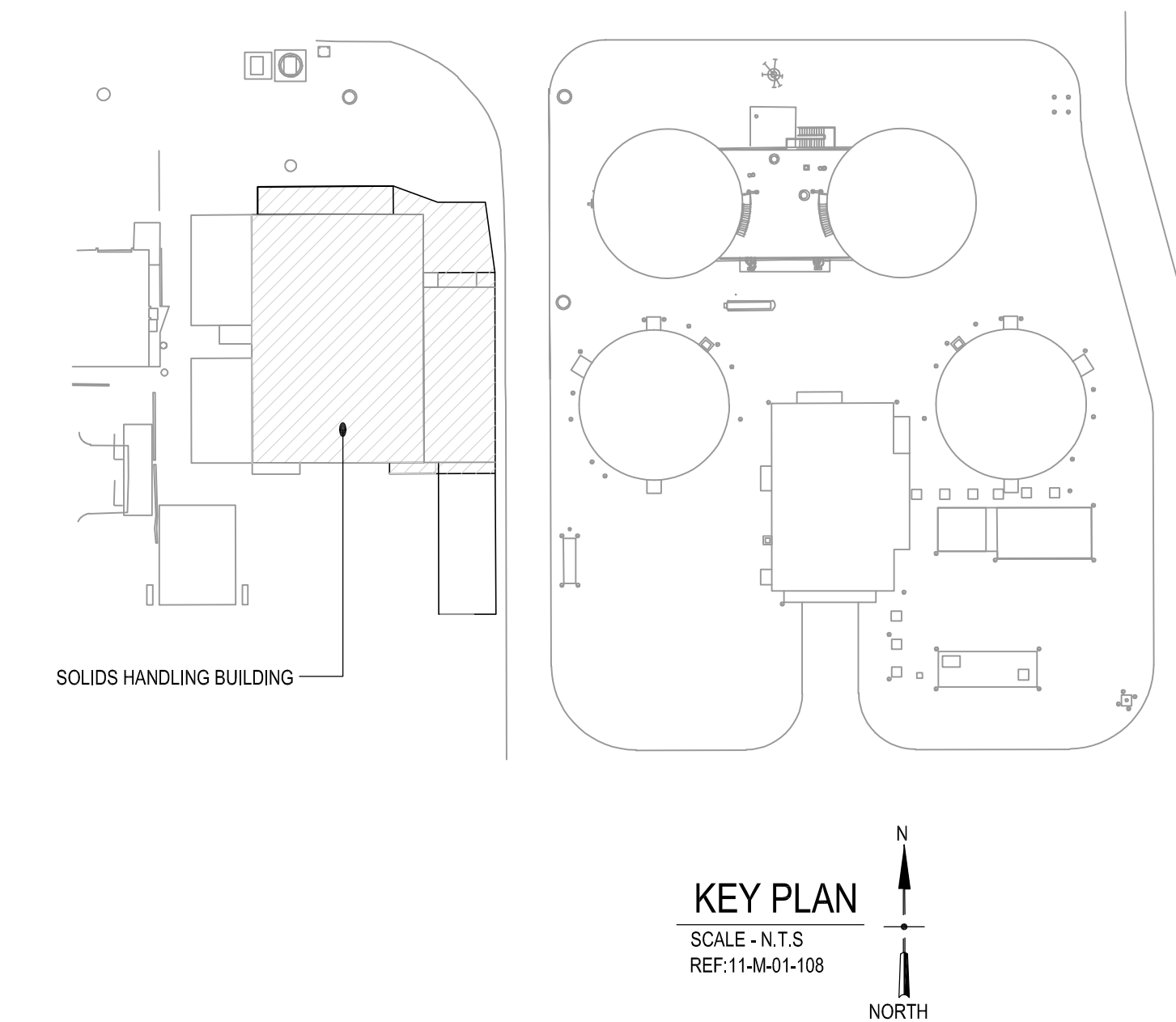
PROJECT INFORMATION

CUSTOMER	-
JOB NAME	HLO WWP
LOCATION	COUNTY OF HAWAII
DESCRIPTION	SOLIDS ODOUR CONTROL BUILDING - PIPE ENCASEMENT DETAILS

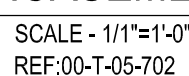
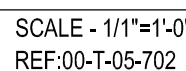
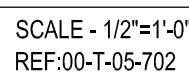
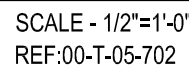
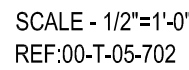


JOB#	2524240001
DRAWING	RP-06

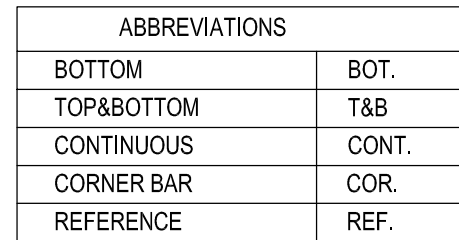
ALL REINFORCEMENTS ARE A706 GR.60 U.N.O.



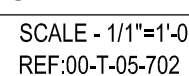
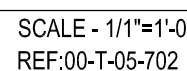
ALL REINFORCEMENTS ARE A615 GR.60 U.N.O



Bar list produced by RebarCAD 2020.0



ENG. PLEASE VERIFY
THE LAP SCHEDULE



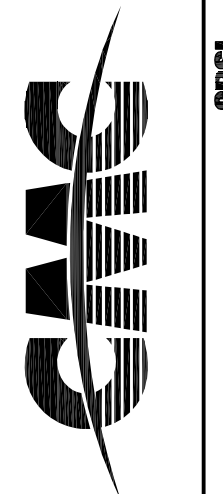
ALL REINFORCEMENTS ARE A615 GR.60 U.N.O

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SUBMITTAL RECORD			
DATE:	06/13/25	FOR:	APPROVAL
DATE:		FOR:	
DATE:		FOR:	
DATE: 08/13/25 - REVISED AS PER MAIL COMMENTS			
DATE:		DATE:	
DATE:		DATE:	

PROJECT INFORMATION	
CUSTOMER -	JOB # 254260023
JOB NAME	DEALER NAME
HILLO WWTP	EMAIL will@masterbar.com
	PHONE (66)-546-6729
LOCATION COUNTY OF HAWAII	
SOLIDS HANDLING BUILDING -- PIPE ENCASMENT DETAILS	
DESCRIPTION:	REF: NOTED



JOB#: 252424000

DRAWING:
RP-08